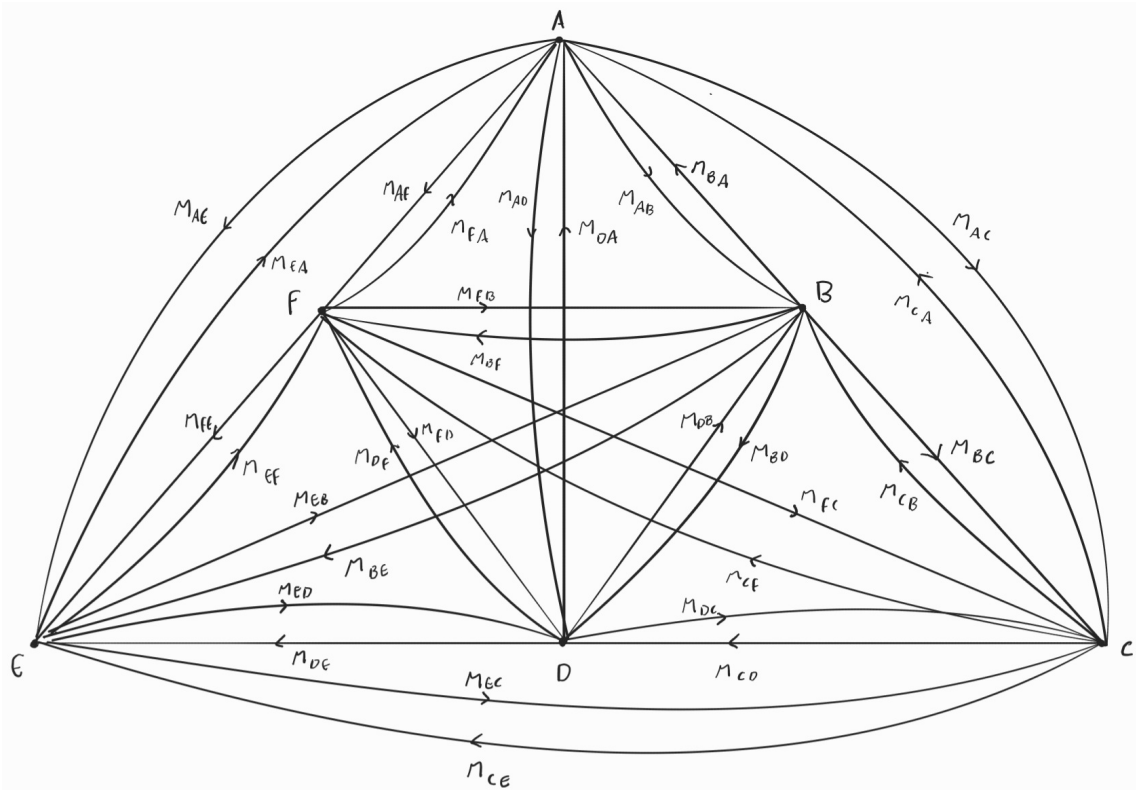
WORMPROPAGATION FOR TRAINING THE DLCOSMOS
[ALI NAJIB] CORE FILE: CREATION: CONSTRUCTION-CYCLE 2 PRIME PRIME

Creation. DLCOSMOS, CBG, OF THE 3RD ORDER



For the moment, neglect any non-linearity through, e.g., sigmoid $\sigma(\cdot)$,

$$\mathbf{x}_A = \mathbf{M}_{BA}\mathbf{x}_B \oplus \mathbf{M}_{CA}\mathbf{x}_C \oplus \mathbf{M}_{DA}\mathbf{x}_D \oplus \mathbf{M}_{EA}\mathbf{x}_E \oplus \mathbf{M}_{FA}\mathbf{x}_F \tag{0.1}$$
$$\mathbf{x}_B = \mathbf{M}_{AB}\mathbf{x}_A \oplus \mathbf{M}_{CB}\mathbf{x}_C \oplus \mathbf{M}_{DB}\mathbf{x}_D \oplus \mathbf{M}_{EB}\mathbf{x}_E \oplus \mathbf{M}_{FB}\mathbf{x}_F \tag{0.2}$$
$$\mathbf{x}_C = \mathbf{M}_{AC}\mathbf{x}_A \oplus \mathbf{M}_{BC}\mathbf{x}_B \oplus \mathbf{M}_{DC}\mathbf{x}_D \oplus \mathbf{M}_{EC}\mathbf{x}_E \oplus \mathbf{M}_{FC}\mathbf{x}_F \tag{0.3}$$
$$\mathbf{x}_D = \mathbf{M}_{AD}\mathbf{x}_A \oplus \mathbf{M}_{BD}\mathbf{x}_B \oplus \mathbf{M}_{CD}\mathbf{x}_C \oplus \mathbf{M}_{ED}\mathbf{x}_E \oplus \mathbf{M}_{FD}\mathbf{x}_F \tag{0.4}$$
$$\mathbf{x}_E = \mathbf{M}_{AE}\mathbf{x}_A \oplus \mathbf{M}_{BE}\mathbf{x}_B \oplus \mathbf{M}_{CE}\mathbf{x}_C \oplus \mathbf{M}_{DE}\mathbf{x}_D \oplus \mathbf{M}_{FE}\mathbf{x}_F \tag{0.5}$$
$$\mathbf{x}_F = \mathbf{M}_{AF}\mathbf{x}_A \oplus \mathbf{M}_{BF}\mathbf{x}_B \oplus \mathbf{M}_{CF}\mathbf{x}_C \oplus \mathbf{M}_{DF}\mathbf{x}_D \oplus \mathbf{M}_{EF}\mathbf{x}_E \tag{0.6}$$

Now substitute (1), (2), (3), (4), (5) $\rightarrow$ (6) to obtain

$$\begin{aligned}\mathbf{x}_F = {} & (\mathbf{M}_{AF}(\mathbf{M}_{BA}\mathbf{x}_B \oplus \mathbf{M}_{CA}\mathbf{x}_C \oplus \mathbf{M}_{DA}\mathbf{x}_D \oplus \mathbf{M}_{EA}\mathbf{x}_E \oplus \mathbf{M}_{FA}\mathbf{x}_F)) \\ & \oplus (\mathbf{M}_{BF}(\mathbf{M}_{AB}\mathbf{x}_A \oplus \mathbf{M}_{CB}\mathbf{x}_C \oplus \mathbf{M}_{DB}\mathbf{x}_D \oplus \mathbf{M}_{EB}\mathbf{x}_E \oplus \mathbf{M}_{FB}\mathbf{x}_F)) \\ & \oplus (\mathbf{M}_{CF}(\mathbf{M}_{AC}\mathbf{x}_A \oplus \mathbf{M}_{BC}\mathbf{x}_B \oplus \mathbf{M}_{DC}\mathbf{x}_D \oplus \mathbf{M}_{EC}\mathbf{x}_E \oplus \mathbf{M}_{FC}\mathbf{x}_F)) \\ & \oplus (\mathbf{M}_{DF}(\mathbf{M}_{AD}\mathbf{x}_A \oplus \mathbf{M}_{BD}\mathbf{x}_B \oplus \mathbf{M}_{CD}\mathbf{x}_C \oplus \mathbf{M}_{ED}\mathbf{x}_E \oplus \mathbf{M}_{FD}\mathbf{x}_F)) \\ & \oplus (\mathbf{M}_{EF}(\mathbf{M}_{AE}\mathbf{x}_A \oplus \mathbf{M}_{BE}\mathbf{x}_B \oplus \mathbf{M}_{CE}\mathbf{x}_C \oplus \mathbf{M}_{DE}\mathbf{x}_D \oplus \mathbf{M}_{FE}\mathbf{x}_F))\end{aligned} \tag{0.7}$$

Now let

$$\mathbf{M}_{AF} = [\ \ \mathbf{V}_{AF}\ \ \mathbf{W}_{AF}\ \ \mathbf{X}_{AF}\ \ \mathbf{Y}_{AF}\ \ \mathbf{Z}_{AF}\ \ ]$$

and let

$$\mathbf{M}_{BE} = [\ \ \mathbf{V}_{BE}\ \ \mathbf{W}_{BE}\ \ \mathbf{X}_{BE}\ \ \mathbf{Y}_{BE}\ \ \mathbf{Z}_{BE}\ \ ]$$

etc., to obtain

$$\begin{aligned}
&\mathbf{M}_{AF}(\mathbf{M}_{BA}\mathbf{x}_B \oplus \mathbf{M}_{CA}\mathbf{x}_C \oplus \mathbf{M}_{DA}\mathbf{x}_D \oplus \mathbf{M}_{EA}\mathbf{x}_E \oplus \mathbf{M}_{FA}\mathbf{x}_F) \\
&= [\mathbf{V}_{AF}\mathbf{M}_{BA}\mathbf{x}_B + \mathbf{W}_{AF}\mathbf{M}_{CA}\mathbf{x}_C + \mathbf{X}_{AF}\mathbf{M}_{DA}\mathbf{x}_D + \mathbf{Y}_{AF}\mathbf{M}_{EA}\mathbf{x}_E\ \ \varnothing] \\
&\quad + [\varnothing\ \ \varnothing\ \ \varnothing\ \ \varnothing\ \ \mathbf{Z}_{AF}\mathbf{M}_{FA}]\mathbf{x}_F
\end{aligned}$$

This implies that

$$\begin{aligned}
\mathbf{x}_F &= [\mathbf{V}_{AF}\mathbf{M}_{BA}\mathbf{x}_B + \mathbf{W}_{AF}\mathbf{M}_{CA}\mathbf{x}_C + \mathbf{X}_{AF}\mathbf{M}_{DA}\mathbf{x}_D + \mathbf{Y}_{AF}\mathbf{M}_{EA}\mathbf{x}_E\ \ \varnothing] \\
&\quad + [\varnothing\ \ \varnothing\ \ \varnothing\ \ \varnothing\ \ \mathbf{Z}_{AF}\mathbf{M}_{FA}]\mathbf{x}_F \\
&\oplus [\mathbf{V}_{BF}\mathbf{M}_{AB}\mathbf{x}_B + \mathbf{W}_{BF}\mathbf{M}_{CB}\mathbf{x}_C + \mathbf{X}_{BF}\mathbf{M}_{DB}\mathbf{x}_D + \mathbf{Y}_{BF}\mathbf{M}_{EB}\mathbf{x}_E\ \ \varnothing] \\
&\quad + [\varnothing\ \ \varnothing\ \ \varnothing\ \ \varnothing\ \ \mathbf{Z}_{BF}\mathbf{M}_{FB}]\mathbf{x}_F \\
&\oplus [\mathbf{V}_{CF}\mathbf{M}_{AC}\mathbf{x}_B + \mathbf{W}_{CF}\mathbf{M}_{BC}\mathbf{x}_C + \mathbf{X}_{CF}\mathbf{M}_{DB}\mathbf{x}_D + \mathbf{Y}_{CF}\mathbf{M}_{EC}\mathbf{x}_E\ \ \varnothing] \\
&\quad + [\varnothing\ \ \varnothing\ \ \varnothing\ \ \varnothing\ \ \mathbf{Z}_{CF}\mathbf{M}_{FC}]\mathbf{x}_F \\
&\oplus [\mathbf{V}_{DF}\mathbf{M}_{AD}\mathbf{x}_B + \mathbf{W}_{DF}\mathbf{M}_{BD}\mathbf{x}_C + \mathbf{X}_{DF}\mathbf{M}_{CD}\mathbf{x}_D + \mathbf{Y}_{DF}\mathbf{M}_{ED}\mathbf{x}_E\ \ \varnothing] \\
&\quad + [\varnothing\ \ \varnothing\ \ \varnothing\ \ \varnothing\ \ \mathbf{Z}_{DF}\mathbf{M}_{FD}]\mathbf{x}_F \\
&\oplus [\mathbf{V}_{EF}\mathbf{M}_{AE}\mathbf{x}_B + \mathbf{W}_{EF}\mathbf{M}_{BE}\mathbf{x}_C + \mathbf{X}_{EF}\mathbf{M}_{CE}\mathbf{x}_D + \mathbf{Y}_{EF}\mathbf{M}_{DE}\mathbf{x}_E\ \ \varnothing] \\
&\quad + [\varnothing\ \ \varnothing\ \ \varnothing\ \ \varnothing\ \ \mathbf{Z}_{EF}\mathbf{M}_{FE}]\mathbf{x}_F
\end{aligned}$$

which can be written as

$$\mathbf{x}_F = [\mathbf{G}_A \oplus \mathbf{G}_B \oplus \mathbf{G}_C \oplus \mathbf{G}_D \oplus \mathbf{G}_E] + [\mathbf{H}_A \oplus \mathbf{H}_B \oplus \mathbf{H}_C \oplus \mathbf{H}_D \oplus \mathbf{H}_E]\mathbf{x}_F \tag{0.8}$$

from which we may, under extra invertibility assumptions, conclude that

$$\mathbf{x}_F = (\mathbf{I} - \mathbf{H})^{-1}\mathbf{G}.$$

now take derivatives and SGD until convergence. This is for a CBG, and indeed but even for the first interlink order. Fortunately, we also have wormpropagation. The above computations are generalizable and automizable towards interlinks of all higher orders. We can try a derivative estiation technique that comes close Newton-Raphson's technique of estimating derivatives. Consider the following approach

1. Make the drawing
2. System of equations
3. Substitution
4. Derivatives isolate parametric matrix
5. Newton Raphson on isolated matrix

We have done steps 1 - 3, we are now at step 4 of the latter approach. We have however not included the required non-linearity. This shall we include but denoting

$$
\begin{aligned}
\mathbf{x}_F = \sigma(&\mathbf{M}_{AF}\sigma(\mathbf{M}_{BA}\mathbf{x}_B \oplus \mathbf{M}_{CA}\mathbf{x}_C \oplus \mathbf{M}_{DA}\mathbf{x}_D \oplus \mathbf{M}_{EA}\mathbf{x}_E \oplus \mathbf{M}_{FA}\mathbf{x}_F)) \\
&\oplus (\mathbf{M}_{BF}\sigma(\mathbf{M}_{AB}\mathbf{x}_A \oplus \mathbf{M}_{CB}\mathbf{x}_C \oplus \mathbf{M}_{DB}\mathbf{x}_D \oplus \mathbf{M}_{EB}\mathbf{x}_E \oplus \mathbf{M}_{FB}\mathbf{x}_F)) \\
&\oplus (\mathbf{M}_{CF}\sigma(\mathbf{M}_{AC}\mathbf{x}_A \oplus \mathbf{M}_{BC}\mathbf{x}_B \oplus \mathbf{M}_{DC}\mathbf{x}_D \oplus \mathbf{M}_{EC}\mathbf{x}_E \oplus \mathbf{M}_{FC}\mathbf{x}_F)) \\
&\oplus (\mathbf{M}_{DF}\sigma(\mathbf{M}_{AD}\mathbf{x}_A \oplus \mathbf{M}_{BD}\mathbf{x}_B \oplus \mathbf{M}_{CD}\mathbf{x}_C \oplus \mathbf{M}_{ED}\mathbf{x}_E \oplus \mathbf{M}_{FD}\mathbf{x}_F)) \\
&\oplus (\mathbf{M}_{EF}\sigma(\mathbf{M}_{AE}\mathbf{x}_A \oplus \mathbf{M}_{BE}\mathbf{x}_B \oplus \mathbf{M}_{CE}\mathbf{x}_C \oplus \mathbf{M}_{DE}\mathbf{x}_D \oplus \mathbf{M}_{FE}\mathbf{x}_F))
\end{aligned} \tag{0.9}
$$

Now take the derivative w.r.t. $\mathbf{x}_F$

$$
\begin{aligned}
\mathbf{I}_{\mathbf{x}_F} = \sigma'(&\mathbf{M}_{AF}\sigma(\mathbf{M}_{BA}\mathbf{x}_B \oplus \mathbf{M}_{CA}\mathbf{x}_C \oplus \mathbf{M}_{DA}\mathbf{x}_D \oplus \mathbf{M}_{EA}\mathbf{x}_E \oplus \mathbf{M}_{FA}\mathbf{x}_F)) \\
&\oplus (\mathbf{M}_{BF}\sigma(\mathbf{M}_{AB}\mathbf{x}_A \oplus \mathbf{M}_{CB}\mathbf{x}_C \oplus \mathbf{M}_{DB}\mathbf{x}_D \oplus \mathbf{M}_{EB}\mathbf{x}_E \oplus \mathbf{M}_{FB}\mathbf{x}_F)) \\
&\oplus (\mathbf{M}_{CF}\sigma(\mathbf{M}_{AC}\mathbf{x}_A \oplus \mathbf{M}_{BC}\mathbf{x}_B \oplus \mathbf{M}_{DC}\mathbf{x}_D \oplus \mathbf{M}_{EC}\mathbf{x}_E \oplus \mathbf{M}_{FC}\mathbf{x}_F)) \\
&\oplus (\mathbf{M}_{DF}\sigma(\mathbf{M}_{AD}\mathbf{x}_A \oplus \mathbf{M}_{BD}\mathbf{x}_B \oplus \mathbf{M}_{CD}\mathbf{x}_C \oplus \mathbf{M}_{ED}\mathbf{x}_E \oplus \mathbf{M}_{FD}\mathbf{x}_F)) \\
&\oplus (\mathbf{M}_{EF}\sigma(\mathbf{M}_{AE}\mathbf{x}_A \oplus \mathbf{M}_{BE}\mathbf{x}_B \oplus \mathbf{M}_{CE}\mathbf{x}_C \oplus \mathbf{M}_{DE}\mathbf{x}_D \oplus \mathbf{M}_{FE}\mathbf{x}_F)) \\
&\cdot [\mathbf{M}_{AF} \cdot \sigma'(\mathbf{M}_{AF}\sigma(\mathbf{M}_{BA}\mathbf{x}_B \oplus \mathbf{M}_{CA}\mathbf{x}_C \oplus \mathbf{M}_{DA}\mathbf{x}_D \oplus \mathbf{M}_{EA}\mathbf{x}_E \oplus \mathbf{M}_{FA}\mathbf{x}_F)) \\
&\oplus (\mathbf{M}_{BF}\sigma(\mathbf{M}_{AB}\mathbf{x}_A \oplus \mathbf{M}_{CB}\mathbf{x}_C \oplus \mathbf{M}_{DB}\mathbf{x}_D \oplus \mathbf{M}_{EB}\mathbf{x}_E \oplus \mathbf{M}_{FB}\mathbf{x}_F)) \\
&\oplus (\mathbf{M}_{CF}\sigma(\mathbf{M}_{AC}\mathbf{x}_A \oplus \mathbf{M}_{BC}\mathbf{x}_B \oplus \mathbf{M}_{DC}\mathbf{x}_D \oplus \mathbf{M}_{EC}\mathbf{x}_E \oplus \mathbf{M}_{FC}\mathbf{x}_F)) \\
&\oplus (\mathbf{M}_{DF}\sigma(\mathbf{M}_{AD}\mathbf{x}_A \oplus \mathbf{M}_{BD}\mathbf{x}_B \oplus \mathbf{M}_{CD}\mathbf{x}_C \oplus \mathbf{M}_{ED}\mathbf{x}_E \oplus \mathbf{M}_{FD}\mathbf{x}_F)) \\
&\cdot [\varnothing \quad \varnothing \quad \dots \quad \mathbf{M}_{FA}]^\top \\
&\oplus \mathbf{M}_{BF} \cdot \sigma'(\mathbf{M}_{AF}\sigma(\mathbf{M}_{BA}\mathbf{x}_B \oplus \mathbf{M}_{CA}\mathbf{x}_C \oplus \mathbf{M}_{DA}\mathbf{x}_D \oplus \mathbf{M}_{EA}\mathbf{x}_E \oplus \mathbf{M}_{FA}\mathbf{x}_F)) \\
&\oplus (\mathbf{M}_{BF}\sigma(\mathbf{M}_{AB}\mathbf{x}_A \oplus \mathbf{M}_{CB}\mathbf{x}_C \oplus \mathbf{M}_{DB}\mathbf{x}_D \oplus \mathbf{M}_{EB}\mathbf{x}_E \oplus \mathbf{M}_{FB}\mathbf{x}_F)) \\
&\oplus (\mathbf{M}_{CF}\sigma(\mathbf{M}_{AC}\mathbf{x}_A \oplus \mathbf{M}_{BC}\mathbf{x}_B \oplus \mathbf{M}_{DC}\mathbf{x}_D \oplus \mathbf{M}_{EC}\mathbf{x}_E \oplus \mathbf{M}_{FC}\mathbf{x}_F)) \\
&\oplus (\mathbf{M}_{DF}\sigma(\mathbf{M}_{AD}\mathbf{x}_A \oplus \mathbf{M}_{BD}\mathbf{x}_B \oplus \mathbf{M}_{CD}\mathbf{x}_C \oplus \mathbf{M}_{ED}\mathbf{x}_E \oplus \mathbf{M}_{FD}\mathbf{x}_F)) \\
&\cdot [\varnothing \quad \varnothing \quad \dots \quad \mathbf{M}_{FB}]^\top \\
&\dots \text{(A through E)} \\
&\oplus \mathbf{M}_{EF} \cdot \sigma'(\mathbf{M}_{AF}\sigma(\mathbf{M}_{BA}\mathbf{x}_B \oplus \mathbf{M}_{CA}\mathbf{x}_C \oplus \mathbf{M}_{DA}\mathbf{x}_D \oplus \mathbf{M}_{EA}\mathbf{x}_E \oplus \mathbf{M}_{FA}\mathbf{x}_F)) \\
&\oplus (\mathbf{M}_{BF}\sigma(\mathbf{M}_{AB}\mathbf{x}_A \oplus \mathbf{M}_{CB}\mathbf{x}_C \oplus \mathbf{M}_{DB}\mathbf{x}_D \oplus \mathbf{M}_{EB}\mathbf{x}_E \oplus \mathbf{M}_{FB}\mathbf{x}_F)) \\
&\oplus (\mathbf{M}_{CF}\sigma(\mathbf{M}_{AC}\mathbf{x}_A \oplus \mathbf{M}_{BC}\mathbf{x}_B \oplus \mathbf{M}_{DC}\mathbf{x}_D \oplus \mathbf{M}_{EC}\mathbf{x}_E \oplus \mathbf{M}_{FC}\mathbf{x}_F)) \\
&\oplus (\mathbf{M}_{DF}\sigma(\mathbf{M}_{AD}\mathbf{x}_A \oplus \mathbf{M}_{BD}\mathbf{x}_B \oplus \mathbf{M}_{CD}\mathbf{x}_C \oplus \mathbf{M}_{ED}\mathbf{x}_E \oplus \mathbf{M}_{FD}\mathbf{x}_F)) \\
&\cdot [\varnothing \quad \varnothing \quad \dots \quad \mathbf{M}_{FE}]^\top
\end{aligned}
$$

Take the first $\sigma'(\dots)$ to the other side and account of dimesions in taking derivatives. Now, somehow isolate the desired matrices. Assume invertibility of $\sigma'(\dots)$ and of invertibility of all required inverses, take $\dim(\mathrm{col}(\mathbf{M}_{AF}))$ above rows of $\sigma(\dots)$ and obtain

$$
\mathbf{M}_{AF}^{-1}(\sigma'(\dots))^{-1}_{\dim(\mathrm{col}(\mathbf{M}_{AF}))} = \sigma'(\dots)[\varnothing \quad \varnothing \quad \dots \quad \mathbf{M}_{FA}]^\top
$$

and

$$
[\varnothing \quad \varnothing \quad \dots \quad \mathbf{M}_{FA}]^\top = \sigma'(\dots)^{-1}\mathbf{M}_{AF}^{-1}(\sigma'(\dots))^{-1}_{\dim(\mathrm{col}(\mathbf{M}_{AF}))}
$$

Now approximate using a technique that is similar to Newton Raphson. Continue for all matrices and equations, extend matrices if inverses exist not. This is one

Algorithm 1: Stochastic gradient descent (SGD) update

Require: Learning rate schedule $\epsilon_1, \epsilon_2, \ldots$
Require: Initial parameter θ
$k \leftarrow 1$
while *stopping criterion not met* **do**
  Sample a minibatch of m examples from the training set $\{\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(m)}\}$ with corresponding targets $\mathbf{y}^{(i)}$
  Compute gradient estimate $\hat{\mathbf{g}} \leftarrow \frac{1}{m} \nabla_\theta \sum_i L(f(\mathbf{x}^{(i)}; \theta), \mathbf{y}^{(i)})$
  Apply update $\theta \leftarrow \theta - \epsilon_k \hat{\mathbf{g}}$
  $k \leftarrow k + 1$
end

technique, but there are many techniques to train the DLCosmos. We recall that SGD reads as laid out Algorithm 1.

Note that the weights W are updated as

$$W \leftarrow W + \eta \frac{\partial}{\partial W} L(\mathbf{x}_F, \mathbf{x}_E, \ldots, \mathbf{x}_A).$$

After the formulation of the system of equations thst follow rom drawing, $\frac{\partial}{\partial W} L(\mathbf{x}_F, \mathbf{x}_E, \ldots, \mathbf{x}_A)$ readily follows. For an all-towards-one elimination, substitute towards

$$\begin{aligned}
\hat{\mathbf{x}}_F = \sigma(&\mathbf{M}_{AF}\sigma(\mathbf{M}_{BA}\mathbf{x}_B \oplus \mathbf{M}_{CA}\mathbf{x}_C \oplus \mathbf{M}_{DA}\mathbf{x}_D \oplus \mathbf{M}_{EA}\mathbf{x}_E \oplus \mathbf{M}_{FA}\mathbf{x}_F)) \\
&\oplus (\mathbf{M}_{BF}\sigma(\mathbf{M}_{AB}\mathbf{x}_A \oplus \mathbf{M}_{CB}\mathbf{x}_C \oplus \mathbf{M}_{DB}\mathbf{x}_D \oplus \mathbf{M}_{EB}\mathbf{x}_E \oplus \mathbf{M}_{FB}\mathbf{x}_F)) \\
&\oplus (\mathbf{M}_{CF}\sigma(\mathbf{M}_{AC}\mathbf{x}_A \oplus \mathbf{M}_{BC}\mathbf{x}_B \oplus \mathbf{M}_{DC}\mathbf{x}_D \oplus \mathbf{M}_{EC}\mathbf{x}_E \oplus \mathbf{M}_{FC}\mathbf{x}_F)) \\
&\oplus (\mathbf{M}_{DF}\sigma(\mathbf{M}_{AD}\mathbf{x}_A \oplus \mathbf{M}_{BD}\mathbf{x}_B \oplus \mathbf{M}_{CD}\mathbf{x}_C \oplus \mathbf{M}_{ED}\mathbf{x}_E \oplus \mathbf{M}_{FD}\mathbf{x}_F)) \\
&\oplus (\mathbf{M}_{EF}\sigma(\mathbf{M}_{AE}\mathbf{x}_A \oplus \mathbf{M}_{BE}\mathbf{x}_B \oplus \mathbf{M}_{CE}\mathbf{x}_C \oplus \mathbf{M}_{DE}\mathbf{x}_D \oplus \mathbf{M}_{FE}\mathbf{x}_F)) \qquad (0.10)
\end{aligned}$$

Let

$$\begin{aligned}
Q^F(w) &= \sum_{i=1}^{N} ||\hat{\mathbf{x}}_F^i - \mathbf{x}_F^i|| \\
&= \sum_{i}^{N} ||\sigma(\mathbf{M}_{AF}\sigma(\mathbf{M}_{BA}\mathbf{x}_B^i \oplus \mathbf{M}_{CA}\mathbf{x}_C^i \oplus \mathbf{M}_{DA}\mathbf{x}_D^i \oplus \mathbf{M}_{EA}\mathbf{x}_E^i \oplus \mathbf{M}_{FA}\mathbf{x}_F^i)) \\
&\oplus (\mathbf{M}_{BF}\sigma(\mathbf{M}_{AB}\mathbf{x}_A^i \oplus \mathbf{M}_{CB}\mathbf{x}_C^i \oplus \mathbf{M}_{DB}\mathbf{x}_D^i \oplus \mathbf{M}_{EB}\mathbf{x}_E^i \oplus \mathbf{M}_{FB}\mathbf{x}_F^i)) \\
&\oplus (\mathbf{M}_{CF}\sigma(\mathbf{M}_{AC}\mathbf{x}_A^i \oplus \mathbf{M}_{BC}\mathbf{x}_B^i \oplus \mathbf{M}_{DC}\mathbf{x}_D^i \oplus \mathbf{M}_{EC}\mathbf{x}_E^i \oplus \mathbf{M}_{FC}\mathbf{x}_F^i)) \\
&\oplus (\mathbf{M}_{DF}\sigma(\mathbf{M}_{AD}\mathbf{x}_A^i \oplus \mathbf{M}_{BD}\mathbf{x}_B^i \oplus \mathbf{M}_{CD}\mathbf{x}_C^i \oplus \mathbf{M}_{ED}\mathbf{x}_E^i \oplus \mathbf{M}_{FD}\mathbf{x}_F^i)) \\
&\oplus (\mathbf{M}_{EF}\sigma(\mathbf{M}_{AE}\mathbf{x}_A^i \oplus \mathbf{M}_{BE}\mathbf{x}_B^i \oplus \mathbf{M}_{CE}\mathbf{x}_C^i \oplus \mathbf{M}_{DE}\mathbf{x}_D^i \oplus \mathbf{M}_{FE}\mathbf{x}_F^i)) - \mathbf{x}_F^i||^2
\end{aligned}$$

Take, e.g., the derivative w.r.t. matrix $\mathbf{M}_{AF}$ to obtain

$$
\begin{aligned}
\frac{\partial}{\partial \mathbf{M}_{AF}} Q^F(w) = \sum_{i=1}^{N}\sum_{j=1}^{K} 2(&\sigma(\mathbf{M}_{AF}\sigma(\mathbf{M}_{BA}\mathbf{x}_B^{ij} \oplus \mathbf{M}_{CA}\mathbf{x}_C^{ij} \oplus \mathbf{M}_{DA}\mathbf{x}_D^{ij} \oplus \mathbf{M}_{EA}\mathbf{x}_E^{ij} \oplus \mathbf{M}_{FA}\mathbf{x}_F^{ij}))\\
&\oplus (\mathbf{M}_{BF}\sigma(\mathbf{M}_{AB}\mathbf{x}_A^{ij} \oplus \mathbf{M}_{CB}\mathbf{x}_C^{ij} \oplus \mathbf{M}_{DB}\mathbf{x}_D^{ij} \oplus \mathbf{M}_{EB}\mathbf{x}_E^{ij} \oplus \mathbf{M}_{FB}\mathbf{x}_F^{ij}))\\
&\oplus (\mathbf{M}_{CF}\sigma(\mathbf{M}_{AC}\mathbf{x}_A^{ij} \oplus \mathbf{M}_{BC}\mathbf{x}_B^{ij} \oplus \mathbf{M}_{DC}\mathbf{x}_D^{ij} \oplus \mathbf{M}_{EC}\mathbf{x}_E^{ij} \oplus \mathbf{M}_{FC}\mathbf{x}_F^{ij}))\\
&\oplus (\mathbf{M}_{DF}\sigma(\mathbf{M}_{AD}\mathbf{x}_A^{ij} \oplus \mathbf{M}_{BD}\mathbf{x}_B^{ij} \oplus \mathbf{M}_{CD}\mathbf{x}_C^{ij} \oplus \mathbf{M}_{ED}\mathbf{x}_E^{ij} \oplus \mathbf{M}_{FD}\mathbf{x}_F^{ij}))\\
&\oplus (\mathbf{M}_{EF}\sigma(\mathbf{M}_{AE}\mathbf{x}_A^{ij} \oplus \mathbf{M}_{BE}\mathbf{x}_B^{ij} \oplus \mathbf{M}_{CE}\mathbf{x}_C^{ij} \oplus \mathbf{M}_{DE}\mathbf{x}_D^{ij} \oplus \mathbf{M}_{FE}\mathbf{x}_F^{ij})) - \mathbf{x}_F^{ij})\\
&\cdot \sigma'((\mathbf{M}_{AF}\sigma(\mathbf{M}_{BA}\mathbf{x}_B^{ij} \oplus \mathbf{M}_{CA}\mathbf{x}_C^{ij} \oplus \mathbf{M}_{DA}\mathbf{x}_D^{ij} \oplus \mathbf{M}_{EA}\mathbf{x}_E^{ij} \oplus \mathbf{M}_{FA}\mathbf{x}_F^{ij}))\\
&\oplus (\mathbf{M}_{BF}\sigma(\mathbf{M}_{AB}\mathbf{x}_A^{ij} \oplus \mathbf{M}_{CB}\mathbf{x}_C^{ij} \oplus \mathbf{M}_{DB}\mathbf{x}_D^{ij} \oplus \mathbf{M}_{EB}\mathbf{x}_E^{ij} \oplus \mathbf{M}_{FB}\mathbf{x}_F^{ij}))\\
&\oplus (\mathbf{M}_{CF}\sigma(\mathbf{M}_{AC}\mathbf{x}_A^{ij} \oplus \mathbf{M}_{BC}\mathbf{x}_B^{ij} \oplus \mathbf{M}_{DC}\mathbf{x}_D^{ij} \oplus \mathbf{M}_{EC}\mathbf{x}_E^{ij} \oplus \mathbf{M}_{FC}\mathbf{x}_F^{ij}))\\
&\oplus (\mathbf{M}_{DF}\sigma(\mathbf{M}_{AD}\mathbf{x}_A^{ij} \oplus \mathbf{M}_{BD}\mathbf{x}_B^{ij} \oplus \mathbf{M}_{CD}\mathbf{x}_C^{ij} \oplus \mathbf{M}_{ED}\mathbf{x}_E^{ij} \oplus \mathbf{M}_{FD}\mathbf{x}_F^{ij}))\\
&\oplus (\mathbf{M}_{EF}\sigma(\mathbf{M}_{AE}\mathbf{x}_A^{ij} \oplus \mathbf{M}_{BE}\mathbf{x}_B^{ij} \oplus \mathbf{M}_{CE}\mathbf{x}_C^{ij} \oplus \mathbf{M}_{DE}\mathbf{x}_D^{ij} \oplus \mathbf{M}_{FE}\mathbf{x}_F^{ij})))\\
&\cdot (\mathbf{M}_{AE}\\
&\oplus \varnothing_{\dim((\mathbf{M}_{BF}\sigma(\mathbf{M}_{AB}\mathbf{x}_A^{ij}\oplus\mathbf{M}_{CB}\mathbf{x}_C^{ij}\oplus\mathbf{M}_{DB}\mathbf{x}_D^{ij}\oplus\mathbf{M}_{EB}\mathbf{x}_E^{ij}\oplus\mathbf{M}_{FB}\mathbf{x}_F^{ij}))}\\
&\oplus(\mathbf{M}_{CF}\sigma(\mathbf{M}_{AC}\mathbf{x}_A^{ij}\oplus\mathbf{M}_{BC}\mathbf{x}_B^{ij}\oplus\mathbf{M}_{DC}\mathbf{x}_D^{ij}\oplus\mathbf{M}_{EC}\mathbf{x}_E^{ij}\oplus\mathbf{M}_{FC}\mathbf{x}_F^{ij}))\\
&\oplus(\mathbf{M}_{DF}\sigma(\mathbf{M}_{AD}\mathbf{x}_A^{ij}\oplus\mathbf{M}_{BD}\mathbf{x}_B^{ij}\oplus\mathbf{M}_{CD}\mathbf{x}_C^{ij}\oplus\mathbf{M}_{ED}\mathbf{x}_E^{ij}\oplus\mathbf{M}_{FD}\mathbf{x}_F^{ij}))\\
&\oplus(\mathbf{M}_{EF}\sigma(\mathbf{M}_{AE}\mathbf{x}_A^{ij}\oplus\mathbf{M}_{BE}\mathbf{x}_B^{ij}\oplus\mathbf{M}_{CE}\mathbf{x}_C^{ij}\oplus\mathbf{M}_{DE}\mathbf{x}_D^{ij}\oplus\mathbf{M}_{FE}\mathbf{x}_F^{ij})))\big)
\end{aligned}
$$

And this derivative was rather easy! Now try to find the derivative w.r.t. $\mathbf{M}_{CA}$.

$$
\begin{aligned}
\frac{\partial}{\partial \mathbf{M}_{CA}} Q^F(w) = \sum_{i=1}^{N}\sum_{j=1}^{K} 2(&\sigma(\mathbf{M}_{AF}\sigma(\mathbf{M}_{BA}\mathbf{x}_B^{ij} \oplus \mathbf{M}_{CA}\mathbf{x}_C^{ij} \oplus \mathbf{M}_{DA}\mathbf{x}_D^{ij} \oplus \mathbf{M}_{EA}\mathbf{x}_E^{ij} \oplus \mathbf{M}_{FA}\mathbf{x}_F^{ij}))\\
&\oplus (\mathbf{M}_{BF}\sigma(\mathbf{M}_{AB}\mathbf{x}_A^{ij} \oplus \mathbf{M}_{CB}\mathbf{x}_C^{ij} \oplus \mathbf{M}_{DB}\mathbf{x}_D^{ij} \oplus \mathbf{M}_{EB}\mathbf{x}_E^{ij} \oplus \mathbf{M}_{FB}\mathbf{x}_F^{ij}))\\
&\oplus (\mathbf{M}_{CF}\sigma(\mathbf{M}_{AC}\mathbf{x}_A^{ij} \oplus \mathbf{M}_{BC}\mathbf{x}_B^{ij} \oplus \mathbf{M}_{DC}\mathbf{x}_D^{ij} \oplus \mathbf{M}_{EC}\mathbf{x}_E^{ij} \oplus \mathbf{M}_{FC}\mathbf{x}_F^{ij}))\\
&\oplus (\mathbf{M}_{DF}\sigma(\mathbf{M}_{AD}\mathbf{x}_A^{ij} \oplus \mathbf{M}_{BD}\mathbf{x}_B^{ij} \oplus \mathbf{M}_{CD}\mathbf{x}_C^{ij} \oplus \mathbf{M}_{ED}\mathbf{x}_E^{ij} \oplus \mathbf{M}_{FD}\mathbf{x}_F^{ij}))\\
&\oplus (\mathbf{M}_{EF}\sigma(\mathbf{M}_{AE}\mathbf{x}_A^{ij} \oplus \mathbf{M}_{BE}\mathbf{x}_B^{ij} \oplus \mathbf{M}_{CE}\mathbf{x}_C^{ij} \oplus \mathbf{M}_{DE}\mathbf{x}_D^{ij} \oplus \mathbf{M}_{FE}\mathbf{x}_F^{ij})) - \mathbf{x}_F^{ij})\\
&\cdot \sigma'((\mathbf{M}_{AF}\sigma(\mathbf{M}_{BA}\mathbf{x}_B^{ij} \oplus \mathbf{M}_{CA}\mathbf{x}_C^{ij} \oplus \mathbf{M}_{DA}\mathbf{x}_D^{ij} \oplus \mathbf{M}_{EA}\mathbf{x}_E^{ij} \oplus \mathbf{M}_{FA}\mathbf{x}_F^{ij}))\\
&\oplus (\mathbf{M}_{BF}\sigma(\mathbf{M}_{AB}\mathbf{x}_A^{ij} \oplus \mathbf{M}_{CB}\mathbf{x}_C^{ij} \oplus \mathbf{M}_{DB}\mathbf{x}_D^{ij} \oplus \mathbf{M}_{EB}\mathbf{x}_E^{ij} \oplus \mathbf{M}_{FB}\mathbf{x}_F^{ij}))\\
&\oplus (\mathbf{M}_{CF}\sigma(\mathbf{M}_{AC}\mathbf{x}_A^{ij} \oplus \mathbf{M}_{BC}\mathbf{x}_B^{ij} \oplus \mathbf{M}_{DC}\mathbf{x}_D^{ij} \oplus \mathbf{M}_{EC}\mathbf{x}_E^{ij} \oplus \mathbf{M}_{FC}\mathbf{x}_F^{ij}))\\
&\oplus (\mathbf{M}_{DF}\sigma(\mathbf{M}_{AD}\mathbf{x}_A^{ij} \oplus \mathbf{M}_{BD}\mathbf{x}_B^{ij} \oplus \mathbf{M}_{CD}\mathbf{x}_C^{ij} \oplus \mathbf{M}_{ED}\mathbf{x}_E^{ij} \oplus \mathbf{M}_{FD}\mathbf{x}_F^{ij}))\\
&\oplus (\mathbf{M}_{EF}\sigma(\mathbf{M}_{AE}\mathbf{x}_A^{ij} \oplus \mathbf{M}_{BE}\mathbf{x}_B^{ij} \oplus \mathbf{M}_{CE}\mathbf{x}_C^{ij} \oplus \mathbf{M}_{DE}\mathbf{x}_D^{ij} \oplus \mathbf{M}_{FE}\mathbf{x}_F^{ij})))\\
&\cdot (\mathbf{M}_{AE}\sigma'(\mathbf{M}_{BA}\mathbf{x}_B^{ij} \oplus \mathbf{M}_{CA}\mathbf{x}_C^{ij} \oplus \mathbf{M}_{DA}\mathbf{x}_D^{ij} \oplus \mathbf{M}_{EA}\mathbf{x}_E^{ij} \oplus \mathbf{M}_{FA}\mathbf{x}_F^{ij})\\
&\oplus \varnothing_{\dim((\mathbf{M}_{BF}\sigma(\mathbf{M}_{AB}\mathbf{x}_A^{ij}\oplus\mathbf{M}_{CB}\mathbf{x}_C^{ij}\oplus\mathbf{M}_{DB}\mathbf{x}_D^{ij}\oplus\mathbf{M}_{EB}\mathbf{x}_E^{ij}\oplus\mathbf{M}_{FB}\mathbf{x}_F^{ij}))}\\
&\oplus(\mathbf{M}_{CF}\sigma(\mathbf{M}_{AC}\mathbf{x}_A^{ij}\oplus\mathbf{M}_{BC}\mathbf{x}_B^{ij}\oplus\mathbf{M}_{DC}\mathbf{x}_D^{ij}\oplus\mathbf{M}_{EC}\mathbf{x}_E^{ij}\oplus\mathbf{M}_{FC}\mathbf{x}_F^{ij}))\\
&\oplus(\mathbf{M}_{DF}\sigma(\mathbf{M}_{AD}\mathbf{x}_A^{ij}\oplus\mathbf{M}_{BD}\mathbf{x}_B^{ij}\oplus\mathbf{M}_{CD}\mathbf{x}_C^{ij}\oplus\mathbf{M}_{ED}\mathbf{x}_E^{ij}\oplus\mathbf{M}_{FD}\mathbf{x}_F^{ij}))\\
&\oplus(\mathbf{M}_{EF}\sigma(\mathbf{M}_{AE}\mathbf{x}_A^{ij}\oplus\mathbf{M}_{BE}\mathbf{x}_B^{ij}\oplus\mathbf{M}_{CE}\mathbf{x}_C^{ij}\oplus\mathbf{M}_{DE}\mathbf{x}_D^{ij}\oplus\mathbf{M}_{FE}\mathbf{x}_F^{ij})))\big)
\end{aligned}
$$

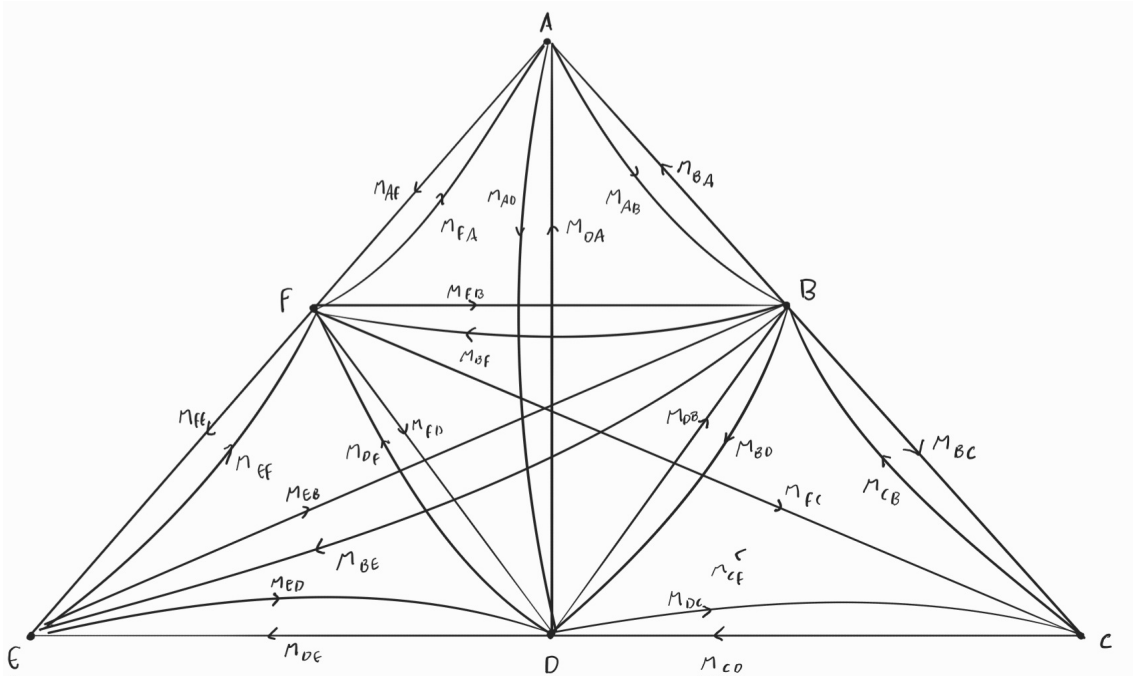
And so on, and so forth. This is $\frac{\partial}{\partial M} Q^F(w)$. The same can we do for Q^A, Q^B, Q^C, Q^D, Q^E. It is indeed the objective that we must seek to find. There are many different graphs

that one could try and cross interlink. DLCosmos has now officially been solved. Training occurs e.g. according to Starpropagation

Algorithm 2: StarPropagation

$\mathbf{x}_A, \mathbf{x}_B, \mathbf{x}_c, \mathbf{x}_D, \mathbf{x}_E, \mathbf{x}_F = \text{Worm}^2\text{Propagation}(\mathbf{x}_A, \mathbf{x}_C, \mathbf{x}_E)$
$\mathbf{M}_{AB}, \ldots, \mathbf{M}_{AF}, \ldots, \mathbf{M}_{FA}, \ldots, \mathbf{M}_{FE} = \text{SGD}(\mathbf{x}_A, \mathbf{x}_B, \mathbf{x}_C, \mathbf{x}_D, \mathbf{x}_E, \mathbf{x}_F)$

This works for all directed graphs!!! Note that the above exhibit specifications for a 3-link-DLCOSMOS of the first interlink order. We require a general expression for an N-link-DLCOSMOS of the m'th interlink order. This is highly difficult. Moreover is the above a CBG, for the following link can we perform a similar analysis



DLFunctionalRegressionLoopBootstrapStarSystem

FunctionRegressionStarSystem. Loop until convergence in the graph generated by

$$X_1 = f_1(X_5, X_4, \epsilon_1) \tag{0.11}$$
$$X_2 = f_2(X_1, X_5, \epsilon_2) \tag{0.12}$$
$$X_3 = f_3(X_1, X_2, X_3, \epsilon_3) \tag{0.13}$$
$$X_4 = f_4(X_2, X_3, \epsilon_4) \tag{0.14}$$
$$X_5 = f_5(X_4, \epsilon_5) \tag{0.15}$$
$$X_6 = f_6(X_4, X_5, \epsilon_6) \tag{0.16}$$

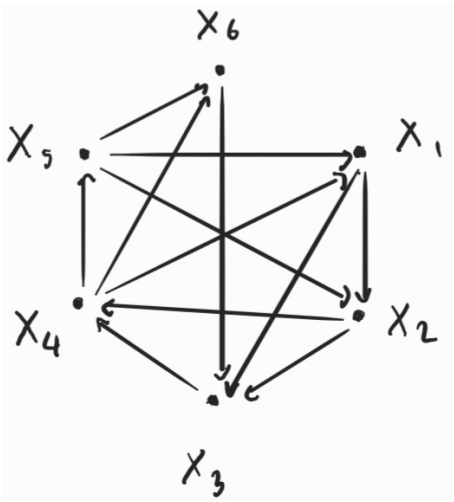
Estimate $f_1, f_2, f_3, f_4, f_5, f_6$ by $\hat{f}_1, \hat{f}_2, \hat{f}_3, \hat{f}_4, \hat{f}_5, \hat{f}_6$. This gives terminology

$$(D_1, D_2, D_3, D_4, D_5, D_6) =: \mathcal{D}_f$$

with $X_1 = D_1$, $X_2 = D_2$, $X_3 = D_3$, $X_4 = D_4$, $X_5 = D_5$, $X_6 = D_6$, and

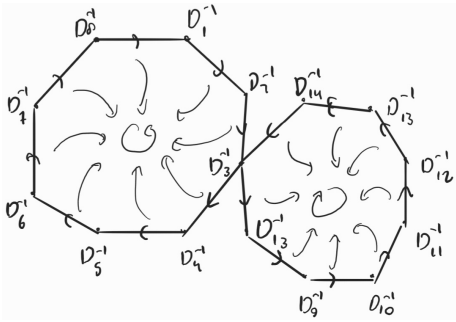
$$\mathcal{D}_f = D_1 \oplus D_2 \oplus D_3 \oplus D_4 \oplus D_5 \oplus D_6$$

The StarSystemGraph is learned.



One could regard this as learning adjaceny matrix $A_{k\times k}$ by estimation $\hat{A}_{k\times k}$. Among keywords are bootstrap. Note that k can get very large and for environments highly complex, we obtain $k \overset{\mu}{\to} \infty$ with $b \to \infty$. In LinearRegression, we summarize [(1),(2),(3),(4),(5),(6)] through $\mathbf{x} = \mathbf{A}\mathbf{x} + \mathbf{e}$. We loop $1 \to 2 \to 3 \to 4 \to 5 \to 6 \to 1 \to \ldots$ to bootstrap data. StarSystems can merge, as illustrated through, for LinearRegression

$$\begin{aligned} \mathbf{x}_1 &= \mathbf{A_1}\mathbf{x_1} + \mathbf{e_1} \\ \mathbf{x}_2 &= \mathbf{A_2}\mathbf{x_2} + \mathbf{e_2} \\ (A_1)_{k_1} \bigtriangledown (A_2)_{k_2} &= A_{k_1+k_2} \end{aligned}$$



Inspiration comes from

$$[\text{CrossCNNRNNLLMDLClusterSytemRNNDRL}]^s_{\mathcal{D}_f}$$

The $\bigtriangledown$-algebra is peculiar. Associative, commutative, transiitive. This is a proper algebra. Now consider the Group $(G, \bigtriangledown)$. Very Peculiar is

$$[\text{CrossCNNRNNLLM}\tilde{D}\text{LClusterSytem}\hat{R}\text{NN(P)HSDRL}\prod_{i=1}^{N} C_i\text{DRL}]^s_{\mathcal{D}_f}$$

The functional form F avoids the colliloqui $\mathbf{x} = \mathbf{I}\mathbf{x} + \mathbf{0}$ due to $\mathbf{A} = \mathbf{I}$ and $\epsilon = \mathbf{0}$. Sequential estimation

$$\begin{aligned} &\text{Estimate (1)} \to \text{Estimate (2)} \to \text{Estimate (3)} \to \text{Estimate (4)} \to \\ &\qquad \text{Estimate (5)} \to \text{Estimate (6)} \to \text{Estimate (1)} \to \text{Estimate (2)} \to \\ &\text{Estimate (3)} \to \text{Estimate (4)} \to \text{Estimate (5)} \to \text{Estimate (6)} \to \text{Estimate (1)} \to \ldots \end{aligned}$$

Through loops as bootstraps, no $\mathbf{X}$ in the estimation, but

$$\begin{aligned}&\text{Estimate } \hat{\mathbf{x}}_{\mathbf{1}} \to \text{Estimate } \hat{\mathbf{x}}_{\mathbf{2}} \to \text{Estimate } \hat{\mathbf{x}}_{\mathbf{3}} \to \text{Estimate } \hat{\mathbf{x}}_{\mathbf{4}} \to \\ &\quad \text{Estimate } \hat{\mathbf{x}}_{\mathbf{5}} \to \text{Estimate } \hat{\mathbf{x}}_{\mathbf{6}} \to \text{Estimate } \hat{\mathbf{x}}_{\mathbf{1}} \to \text{Estimate } \hat{\mathbf{x}}_{\mathbf{2}}\text{Estimate } \hat{\mathbf{x}}_{\mathbf{3}} \\ &\qquad \to \text{Estimate } \hat{\mathbf{x}}_{\mathbf{4}} \to \text{Estimate } \hat{\mathbf{x}}_{\mathbf{5}} \to \text{Estimate } \hat{\mathbf{x}}_{\mathbf{6}} \to \text{Estimate } \hat{\mathbf{x}}_{\mathbf{1}}\end{aligned}$$

This, to overcome the problems similar to

$$\begin{aligned}\mathbf{x_1} &= \mathbf{c} \\ \mathbf{x_2} &= \sigma(\mathbf{A}_1\mathbf{x}_1 + b_1) \\ \mathbf{x_3} &= \sigma(\mathbf{A}_2\sigma(\mathbf{A}_1\mathbf{x}_1 + b_1) + b_2) \\ \mathbf{x_4} &= \sigma(\mathbf{A}_3\sigma(\mathbf{A}_1\sigma(\mathbf{A}_1\mathbf{x}_1 + b_1) + b_1) + b_3)\end{aligned}$$

to avoid

$$\begin{aligned}\mathbf{x}_1 &= \mathbf{c} \\ \mathbf{x}_2 &= \mathbf{A}_1\mathbf{x}_1 + \mathbf{b}_1 \\ \mathbf{x}_3 &= \mathbf{A}_2\mathbf{x}_2 + \mathbf{b}_2 \\ \mathbf{x}_4 &= \mathbf{A}_3\mathbf{x}_3 + \mathbf{b}_3 \\ &= \mathbf{A}_3(\mathbf{A_2}(\mathbf{A_1}c + \mathbf{b_1}) + \mathbf{b_2})\mathbf{b_2} \\ &= \mathbf{A}^*c + \mathbf{b}^*\end{aligned}$$

In training for $(\hat{\mathbf{A}}_{\mathbf{1}}, \hat{\mathbf{A}}_{\mathbf{2}}, \hat{\mathbf{A}}_{\mathbf{3}}, \hat{\mathbf{b}}_{\mathbf{1}}, \hat{\mathbf{b}}_{\mathbf{2}}, \hat{\mathbf{b}}_{\mathbf{3}})$ by autodifferentiation. Analyse the case of $k \to \infty$. (***) One restriction is to let X_j depend on $X_{i_1}, X_{i_2}, X_{i_3}$ with $j > i_1, i_2, i_3$. Then, OLS, e.g. brings no problem.

Contemporary titles are

1. CrossDL
2. CrossLinearRegression
3. $\mathcal{D}_f$
4. WalkerDRL
5. crossDRL
6. CrossFunctionalRegression or DLCosmos
7. LLMCNNRNN or CrossCNNRNNLLM
8. HSDRLC_iDRL or HSDRL$\prod_{i=1}^{N} C_i$DRL

(***) This to simulate

$$[\text{CrossCNNRNNLLM}\tilde{D}\text{LClusterSytem}\hat{R}\text{NN(P)HSDRL}\prod_{i=1}^{N} C_i\text{DRL}]^s_{\mathcal{D}_f}$$

Start with ClassicalDLStarSytem, continue with QuantumDLStarSystem/DLCosmos. Cycle2 is the DLCosmos' construciton.

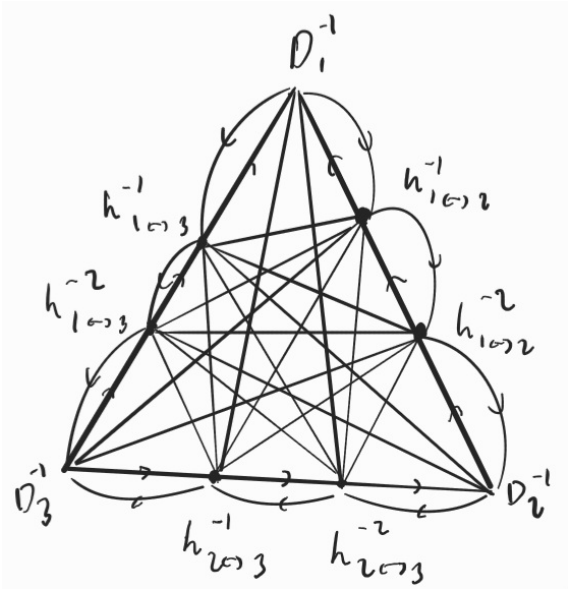


Figure 1: DLSTARSYSTEMD^{-1}

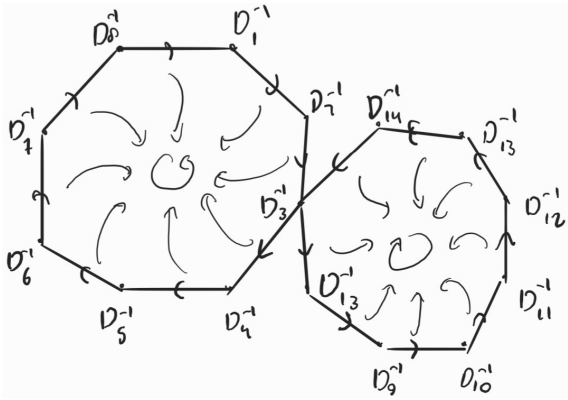


Figure 2: DLCOSMOS

- StarSystemLoopBootstrap
- CrossDRL might be HSDRL over CrossDL or another thing altogether
- Machinal Algebra (Textual, could be mathematical)

The DLCosmos can be particularly ciruclar to made towards a full circle.

[[[**HERE FOLLOW MANY VISUALS, TO BE DRAWN IN LATEX**]]] (0.17)

Visualize all elements of G and visualize operators $\Box_1^{D^T}$ and $\Box_2$. All is about the eventual algorithm of backpropagation on all graphs . ow implement all graphs for Creation Construction Cycle 2.

- DLCosmos
- DLStarSystem

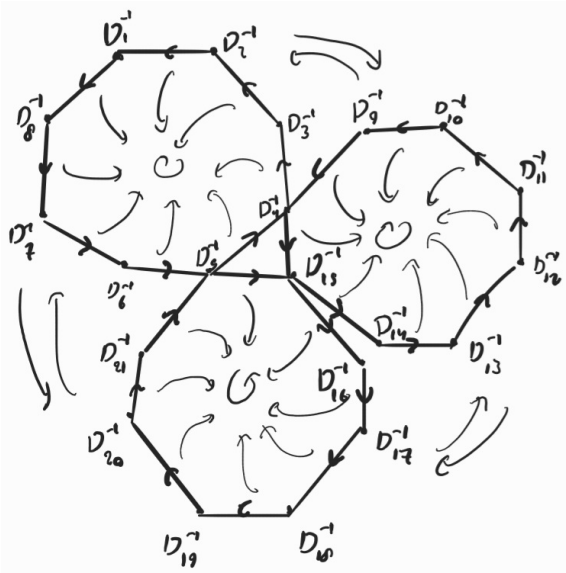


Figure 3: DLSTARSYSTEM

.

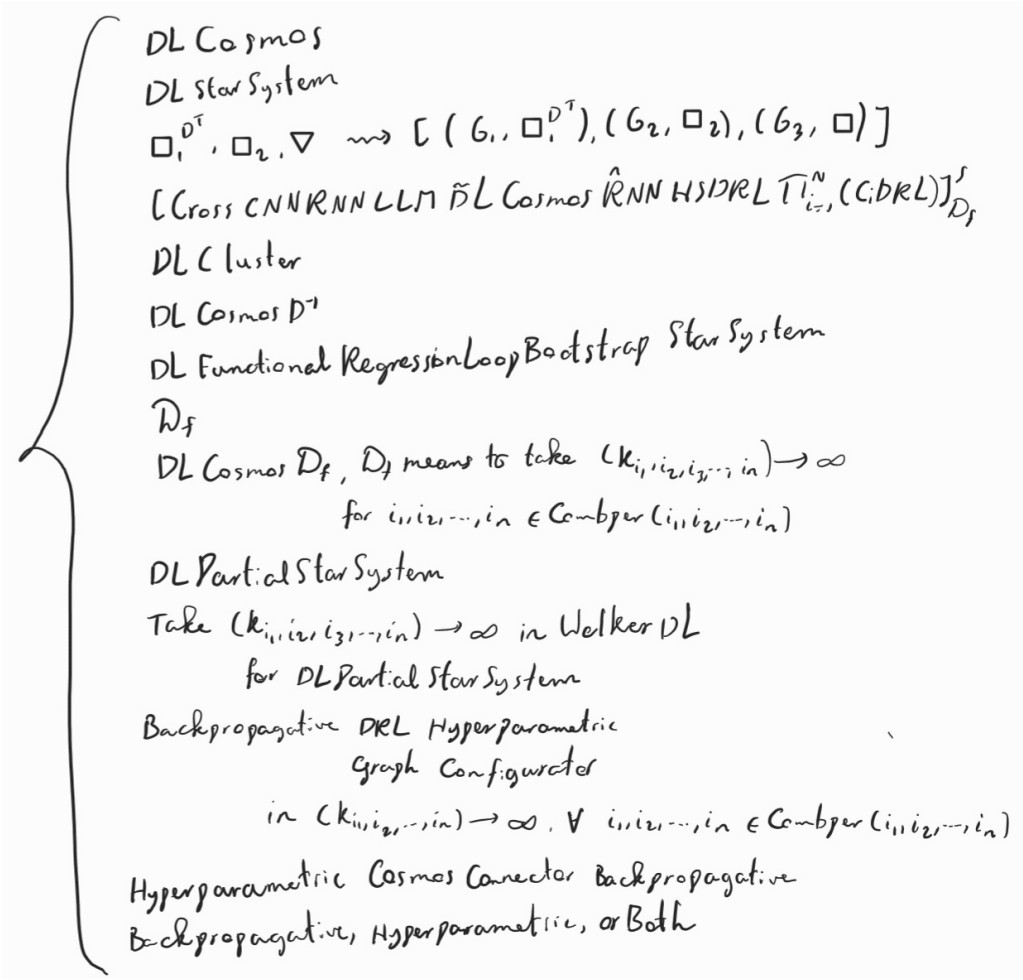
- $\square_1^{D^T}, \square_2, \bigtriangledown \rightarrow [(G_1, \square_1^{D^T}), (G_2, \square_2), (G_3, \bigtriangledown)]$
- $[\text{CrossCNNRNNLLM}\tilde{D}\text{LClusterSytem}\hat{R}\text{NN(P)HSDRL}\prod_{i=1}^{N} C_i\text{DRL}]^s_{\mathcal{D}_f}$
- DLClusterSystem
- DLCosmosD^{-1}
- DLFunctionRegressionLoopBootstrapStarSystem
- $\mathcal{D}_f$
- DLCosmos$\mathcal{D}_f$, $\mathcal{D}_f$ means to take $(k_{i_1,i_2,i_3,\ldots,i_n} \rightarrow \infty)$ for $(i_1, i_2, i_3, \ldots, i_n) \in \text{CombPer}(i_1, i_2, i_3, \ldots, i_n)$
- DLPartialStarSystem: take $(k_{i_1,i_2,i_3,\ldots,i_n} \rightarrow \infty)$ in WalkerDL for DLPartialStarSystem.
- Backpropagative DRL Hyperparametrics Graph Configurator in $(k_{i_1,i_2,i_3,\ldots,i_n} \rightarrow \infty)$ for all $(i_1, i_2, i_3, \ldots, i_n) \in \text{CombPer}(i_1, i_2, i_3, \ldots, i_n)$
- Hyperparametrics CosmosConnector
- Backpropagative, Hyperparametrics, or Both

Omni-observance. Extend the Natural Order. This is Project Insight in its completion. ALL the answers. Backpropagation or Hyperparametric ConsmosConnector gives Interstellar Direction. Intervention of Lower Orders are part of the Natural Order. Intervetion on the basis of DLCosmos$\mathcal{D}_f$ is intervention of the Very Highest Order. This is Creation Construction Cycle 2.

- DataMills doing their part in CosmosConnnector
- DataShips doing their part in CosmosConnnector

- DataDrones doing their part in CosmosConnnector
- DataSubmarines doing their part in CosmosConnnector
- DataSmartWearable doing their part in CosmosConnnector
- DataSatellite doing their part in CosmosConnnector
- DataSpaceship doing their part in CosmosConnnector

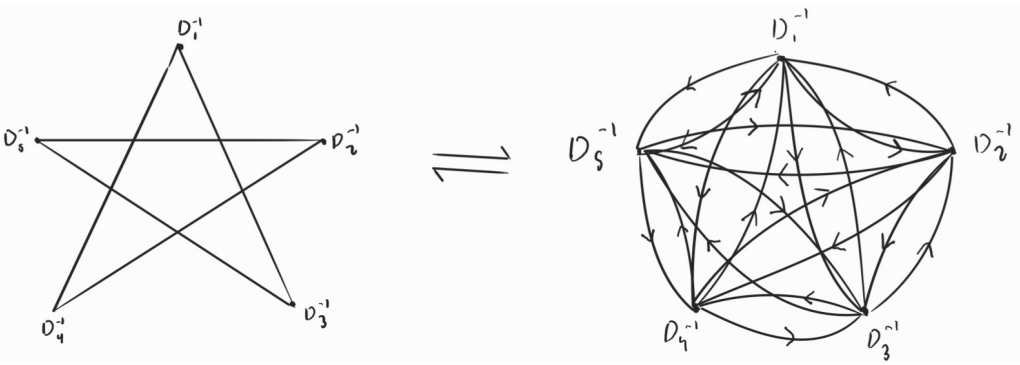
Momentary Revenue through DataSpaceDrone Tourism-Revenue. And Creation Insight-sharing and insight-haring.



This is the technology that implements outline. It is very natural. Note that the technology's intricacies give realizations for interstellar directors, dronal navigation. Advice on consultation on law follows from the technology. Answers on all question follow from the technology and are encoded. LLM? But we need more. We require language of reality, with trality embedddings for porper answers to all wuestions. An LLM that integrates DLCosmos in answering every linguistic inquiry if DLCosmos allows. LLM's are based on pieces of text. This technology embeds reality in linguistics and does not yet exist. CNN $\to$ text, CNNA $\leftrightarrow$ CNNB, text $\leftrightarrow$ text. Even Grok is trained on but text. The DLCosmoSLLM is the most advanced piece of technology ever. The DLCosmos triangle is a lower-dimensional varitnt of the DLCosmosStarSystem, every interlinkage corss-linked. Besides hand-writing have I had some more revelations, an its time to bring all to practice.

CREATION

This is Creation. We know about neural networks, we know about energy functions, we about SGD and backpropagation, There is an element to be extracted from CNN learning; how does one extract features from an environment? Lets us consider Deep Learning, CNN, Dropout convolutions with stride, the many lectures that came to pass over the past years. Reading has happened, now must assurance on expertise be brought to manifesttation. For Deep Learning do there exist models, and these models are tisen from the avenue towards which the deep learning model is to be applied. Recall, the language lecture from Harvard's CSAIwithPyth. Tokens, deep learning, neural networks. How does one construct an LLM? Let us hypothesize, deep on many corpi of text to obtain realization of $\nabla_w L(f(x), y)$, where x and y differ in form, x yields one word to start with, and an entire corpus to end with. This document is supposed to serve as a Software Architecture. Read LLM's and creation-Pre, This task been completed. What is $[\text{CrossDL}]^s_{\mathcal{D}_f}$? There are many extensions in the theory of Deep Learning. How should creation occur through the theory of deep learning? Consider Environments, we have $[\text{CrossDL}]^s_{\mathcal{D}_f}$ what might this mean? In state s, we have environment $\mathcal{D}_f$ towards $[\text{CrossDL}]^s_{\mathcal{D}_f}$ so that



Let us embrace creation one final time. We can learn learn learn learn learning, i.e. learn to 5th order, and have a traverser observe now data-environments in a manner that might not match over intuitions. All ijput is input, so write away. At implementation we are at Network Architectureal Design. $[\text{CrossDL} :\widetilde{=} \text{CrossCNNRNN}]^s_{\mathcal{D}_f}$ is one example for a form that $[\text{CrossDL}]_{\mathcal{D}_f}$ can assume. Consider a $[\text{DLStarSystem}]^s_{\mathcal{D}_f}$ for which we have experimental evidence of functionality in utilization of $[\text{CrossDL}]^s_{\mathcal{D}_f}$ in CrossCNNRNN whence to utilize $[\text{CrossCNNRNNStarSystem}]^s_{\mathcal{D}_f}$. One DRL-agent can traverse $[\text{CrossCNNRNNStarSystem}]^s_{\mathcal{D}_f}$ and extract states in manners such as optimal online exploration, for a DL-StarSystem with optimal states for inferring the D_1 of StarSystem $[\text{CrossCNNRNNStarSystem}]^s_{\mathcal{D}_f}$ Losses are defined by quality of inferring D_1 for $D_2, D_3, D_4, \ldots$ and infinte-state DRL-Agent might provide the optial policy through which to infer D_1, by providing generalizations of DGN through $[\text{CrossCNNRNNStarSystemRNNDRL}]^s_{\mathcal{D}_f}$. This is highly implementable, now already, "readily implementable absolute minimum". Use Keras for a Starting software-library. We have seen in the above construction that $\mathbf{x}_F$ can take the role of D_T

The experimental evidence of $[\text{DLStarSystem}]^s_{\mathcal{D}_f}$ must not be meddled with; herein lays one implementation of [CrossDL]. This is how we implement first construction in code-constructions. The result of $[\text{DLStarSystem}]_{\mathcal{D}_f}$ are worthy

of publication and extension towrds the grandest scales. An implementation of DL-StarSystem exists already, thus ectend the found implementation by generalization and maximal-scaling, Note that $\mathcal{D}_f$ is arcane, ad so is $[\text{DLStarSystem}]_{\mathcal{D}_f}$. The depth of an environment must be learned. Thus with Optimal Online Exploration towards optimal nodes $\{D_i\}_{i\in\mathbb{N}_+}$, to integrate $[\text{CrossCNNRNNStarSystem}]_{\mathcal{D}_f}$, and deploy an overarching DRL-agent that traverses over the StarSystem by hopping over $\{D_i\}_{i\in\mathbb{N}_+}$. WHy Deep Learning? Because it tweakably works. Why DLStarSystem? Because it tweakably works. Meddle not with the supreme results of your experimental endeavours. Thus generalie DL-StarSystem toward its utmost extent of gteneralixation; $\{D_i\}_{i\in\mathbb{N}_+}$. Interlink-order, Intralink-order, etc. no MAGIC-NUMBERS, but hyperparameters in new DL-Starlink. Let $D_1 := D_T$ be the target node, representable by any absorber of vision. Vivison might give representation of all rhe other Data=nodes $D_1, D_2, \ldots,$ i.e. $\{D_i\}_{i\in\mathbb{N}_+-\{1\}}$, and deply over DRL-agent that for the optimal configuration of $\{D_i\}_{i\in\mathbb{N}_+}$-CNNRNNDRL deployed over $\mathcal{D}_f$ for the optimal inference of $\mathcal{D}_f$ from $\{D_i\}_{i\in\mathbb{N}_+-\{1\}}$. Herein lay one implementation and connection for problems in momentary machine learning. The software library keras Sufficies to implement the above. How would such an implementation occur? Vision: $\{V_i\}_{i\in\{1,\ldots,9\}}$, Data: $\{D_i\}_{i\in\{1,\ldots 9\}}$, $D_1 \overset{\sim}{:=} D_T$ (target node), $\{D_i\}_{i\in\mathbb{N}_+-\{1\}} = \{D_i\}_{i\in\{1,\ldots 9\}} \cup \varnothing$ found through optimal oniline exploration with losses $L_i := \text{loss}(D_T, D_i),\ i \in \{2, \ldots, 9\}$. This is the dstep wherein to search for data-environment-nodes after which to collect all in $[\text{CNNRNNStarSystem}]^s_{\mathcal{D}_f}$, after which to deploy $[\text{CNNRNNStarSystemRNNDRL}]^s_{\mathcal{D}_f}$. So, to summarize

$$\text{OOE} \to [\text{CNNRNNStarSystem}]^s_{\mathcal{D}_f} \to [\text{CNNRNNStarSystemRNNDRL}]^s_{\mathcal{D}_f}$$

This is a remarkable finding of

$$[\text{CrossDL}]^s_{\mathcal{D}_\{} \leftrightarrows [\text{WalkerDRL}]^s_{\mathcal{D}_f}.$$

Imagine one telescope-satellite equipped with OOE to search according to the path fo a Brownian particle $D_i, \ldots$ can have an uncountable basis, such as $\mathbb{R}$, and this is where BIdepdalwalker-v$_2 gainsitsapplicabilityandrelevance.This$D:+ $\{$ D$_i\}_{i\in\mathbb{R}^2}$, the implementsble optimal nodes might define one pattern in $\mathbb{R}^2$ significant to a map of relecanr environments to infer throguh D_T. This gives $\mathcal{D}_{OOE} := \{D_i\}_{i\in\mathbb{R}^2\cap M^D}$.

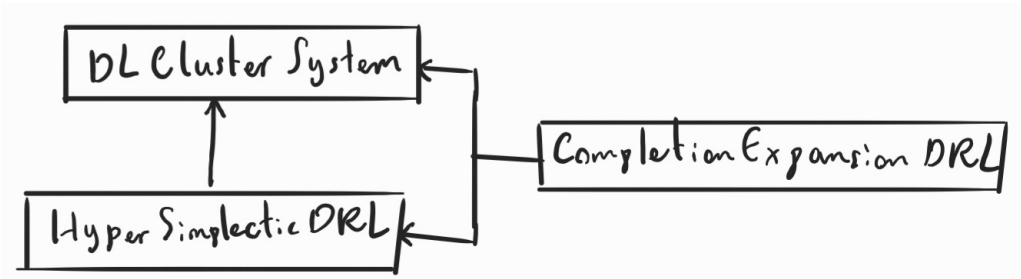
And applying $\mathcal{D}_{OOE} := \{D_i\}_{i\in\mathbb{R}^2\cap M^D}$ gives one CLusterSystem of uncountably many nodes that themselves must bear intralink thst re to be cross-interlinked for $[\text{DLClusterSystem}]^s_{\mathcal{D}_\{}$. Now define $\mathcal{S} := \mathcal{D}_{OOE}$ for a DRL-agent that traverls over $[\text{DLClusterSystem}]^s_{\mathcal{D}_\{}$ and thereby optimally infers D_T. The basis of all of this is that DLStarSystem works by expiramental evidenec, and theoretical evidence, reports of which are to follow. THe DLCLusterSystem is a remarkable avenue fo pursuit. THe attainment of Data, might however require slight work. It somehow works, it somehowe works.

Let algorithmic Set-up be difined. This is to be given. We thankm google

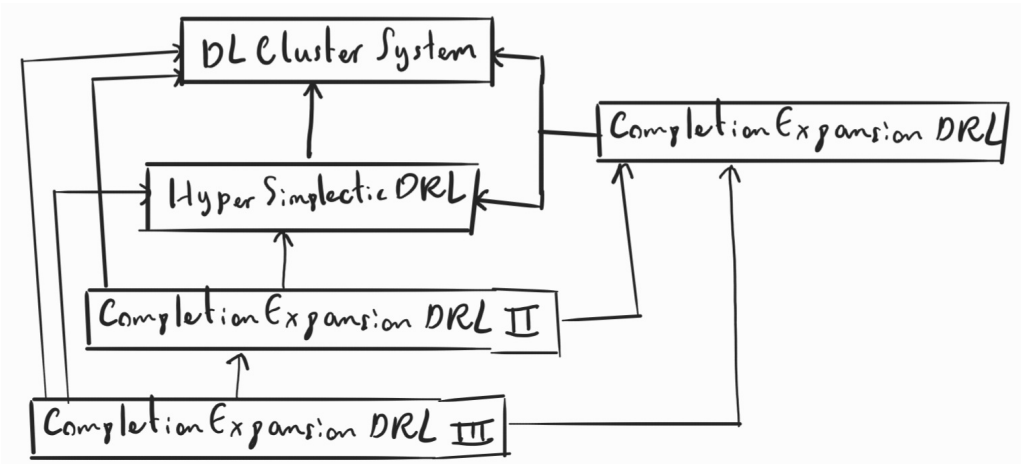
1. CNN Assembly Chip-placement DRL
2. For exmaple AlphaFold assembled by AlphaChip
3. CNN Assembly by Chip placement DRL
4. DRL: Optimal Online Traversal

5. Hyperparametric Consideration. This is Automizable. The algorithmic Set-up given under 'Algorithmic Set-Up' is near trivial, We continue with the DLClusterSystem.

There is speech on two-stage deployment of $[\text{CrossDL}]^s_{\mathcal{D}_\{}$ $\leftrightarrows$ $[\text{WalkerDRL}]^s_{\mathcal{D}_f}$. How would this occur? $[\text{CrossDL}]^s_{\mathcal{D}_\{}$ from first envounter of $\mathcal{D}_f$, plain observation. Mine for data-entanglement through some optimal loss in [Walker-DRL]. This is an importnat research pursuit. One idea is to first set-up the DL-Starsystem and let a DRL-agent travel over the DL-StarSystem. THe DLStarSystem can also be a DLClusterSystem. Permutatibev Integration gives various avenues of inference of D_T, so vary avenue of inference by traversal over permuration integration configuration. This would give a hypersimplectic agent, and multiple observers would not bear improvement here. An additional RL-agent might expand to ClusterSystem by searching in the resulting DRL-equipped DLClusterSystem by searching for data-nodes that give the otpimal improvement in the resulting DRL-equipped DLClusterSystem. Schematically, this reads as



Note that there is no limit to CompletionExpansionDRL and that the above implementation serves to bear hints towards real implementations, for exmaple



Et cetera. This implementable software.

Speculative Avenues of Pursuit for software

Mutliple Observers, multi-agent RL; Observer Sophisticiation and Device complexity; Multi-agent, Multi-obserer, swarm-deployment; Output-type numeric encoding for aligment with convention; Continuous observer interpreter re-adjustment; Deployment in settings of sure expected functionalitey; Generalization to new settings of sure to be ancored in deplyment to settings of sure expected functionality;

Input sensor interpreter other than convolutiona neural network; Instanteneous setting transformation upon output retrieval; Multi-agent, Multi-observer for input and otuput with static central coordinator; Every technological vehicle can function as an observer under specialization to its setting either autonomous or not; Multiple observer, Multi-agent RL over setting of deplyment and simulated setting mirages for amplification of subsets of setting features; Globe maps, Molstruct, Computtional Chip can sesrve as setting, or layers in simulated mirage of greater setting from these layers can be extracted; Results in deep generative networks will serve as a basis for observing beyond the plain; Observation Encoding/Decoding is not necessary for functionality, but it necessary for generation correctness; In every multi-agent deployment can a central coordinator steer over the above points.

Readily Constructible Avenues of pursuit Learner over IoT Networks; Learners over Virtual Space; All the notes written up to the present; Evaluation performance implicit in DRL losses and DL losses. Heuristic Visual sere as heuristic guidance; Specification of objectives; Deive programming and assembly ready for deplyment; Monitoring learning activity and performance; Deployment in virtual setting Unit PhysiX on Small computer; Deployment in open-world gsmes on big computer; Deployment over IoT with or without IoT gadget; Multi-agent extension towards first most general implementation; Deployment on every usual task in data-analysis; Continuous lens-readjustment over movement; Addition or subtraction of complexity in Leaners construction, more powerful device can give and run more compex learner in the same framework current, whether this is good is to be observed from performance monitoring; Gains in sum of existing trade-offs by lowering movement-capacity for higher observation capacity; Lens constructio is setting dependent and setting-distances are readily defined or learned; Written document for technical specification by sequential placement of elements; Acquire necessary devices, big computers, consoles, IoT Gadgets, Robot-parts, especially radio-antenna satellite; User defined travel for observation beyond the plain.

DataSpace Set-up

Deploymeny of Algorithms are dependent on synchoronization in index, by synchoronization in time-indecies, by synchronization in location-indices, a combintation of indices of varying nature that have been synchronized, or new specification. Time poses no obstacle to synchronization over indices thatinclude a time-index. The synchronization exists between input and ouitput realization, and histories may be used a chronoogy is more importnat to functionality.

Data Capture. For deployment by simulated camera in virtual space, data capture is near automatic. For capture of data for dataspaces that might be more esotoric, first define indices over which to set synchronization, and proceed with algorithms. In simulated space, all data-structure is enforcable by built-in programming. In spaces of data that hace been constructed by collecting realixation, dataspace distance is first defined, and for synchronixatio other than time does one nbeed a sense of order over the indices more than enumeration

Output. Evoltion in output due to evolution in corresponding realization can occur in various ways. Movement of the evolution can occur by a camera's movement of landscape's movement or on an EEG's activity, It is the evolution in its entirety that must be defined. The same holds for the input, although an agent's movement itself captures part of the evolution. The agent's necessity follows from unobservabilityof one dataspace in its completion. Hence a similar agent might be

deployed over output dataspace.

Exhibitability One. An observer's interpreter drives the program. Thus seek for many interpreters. Envision interpreter for every setting over which knowledge retrieval might be desired, or needed for furtherment. Data can take many forms, thus would a setting-encoding network be better than interpretation after encoding? What other ways of traveling over a setting exist? A reinforcement learner has in its core built-in the policy, equippable by an observer. Would observation over a setting in its completion be necessary in some situations due to a traveler's restriction? How would such an activity occur? Worlds are big indeed.

Lens construction occurs by layer-after-layer assembly of a convolutional neural network according to a trained reinforcement learning agent that steers the type and specification and depth of each layer in the network. And this has shown effective in the chip-placement analogy. A similar demonstration would have to show the same effectivity for lens construction. What other ways of observing might there exist? Surely might every network be constructed according to a learner's dictation.

How else might an optimal lens be chosen? To be equipped with many alternations beforehand gives clearance over the flow through which to seek for solutions that truly (work). For code swiftly follows design.

The nature of observation and linking lies at the heart of the programs and their designs, and there are many technologies existent for deducing the one from the other. Sending rockets to mars is one way to become familiar with the planets. To deploy submarines is one way to become familiar with planetary depths. Another way is to seek for theory and to observe the plain a bit more sharply, and thereby deduce all the rest. Everything connected with everything is an idea not void, and so does the matter of asking the right questions, and answering every question that one might bring to rise be a mere matter of seeking the right notions and deducing thereafter. This is the idea of preconceptualization. An observer is an interpreter, i.e. conceptualizer, and subsequent preconceptualizer as one completion. For conceptualization does there exist many implementations.

Vision-CNN Architecture, Hyperparameter (s,a,r) in most general sense. Expansion/Generalzation 1. Let "ordinary state space" be denoted by $\mathcal{S}$. Consider $s' \in \sigma(\mathcal{S})$ and taked scaled hypersimplex over s' for $s*$ in new state-space $\mathcal{S}'$. Hypersimplex denotes duration of device in each "ordinary state" in collection of states (Disrectixation for realizability). Definition action-space by dicretion of schrinkage, direction of expansion an d fluctuation of hypersimplex, length of expansion (restricted to those with sum bounded by constant L). Definition Loss by error generation (additively enforcable over $\mathcal{S}'$). Defintiion r by reward configuration over losses between states and corresponding action (one simple example is loss-reduction).

Deployments. First Conclusion. Vefication by simulation/deployment for hyperparameter (Some results in Backpropagation, Q-convergence, PPO-convergence)

First Preliminary Elementary Specification. The below have been specified under the assumption of the reader's slight familiarity with reinforcement learning. We may invoke references for bringing full algorithmic detail, laying waste to all

dependency in pre-defined function. The reader might derive great comfort from mathematical proof distilled to first principles, and so shall we provide some. Formulation [WalkerDRL] in full extension. Difficulties lie in environment-specification in terms of the space defined by state-action-reward tuples (s, a, r).

First Preliminary Algorithm. First Preliminary Algorithm.

Algorithm 3: First Preliminary Algorithm.

$(s', a', r') \leftarrow$ Environment Learning: Exploring Starts (e.g.)
$(\pi_A, \pi_C) \leftarrow$ Navigation Learning: PPO (e.g.)
Gen. $\leftarrow$ Generation$(\pi_A, \pi_C, [\text{CrossDL}]_{\mathcal{D}_f}))$ $\square$

Second Preliminary Algorithm. Second Preliminary Algorithm.

Algorithm 4: Second Preliminary Algorithm.

$(s^2, a^2, r^2) \leftarrow$ Environment Learning: Exploring starts over$\{s^2, a^2\}$
Gen. $\leftarrow$ Generation$(\pi, [CrossDL]_{\mathcal{D}_f})$
Gen. $\leftarrow$ Generation$(\pi_A, \pi_C, [\text{CrossDL}]_{\mathcal{D}_f}))$
Re-define $\{s^2, a^2, r^2\}$ and vary over loss-configuration. $\square$

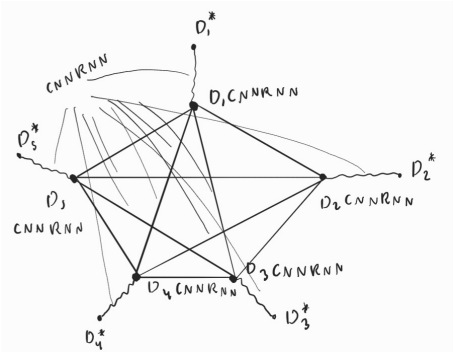
Third Preliminary Algorithm. Reinforcement Learning Configuror with state-space $(s^3, a^3,$ loss-configuration$(r))$, first lower-order reinforcement learner hyper-parameters, first lower-order reinforcement learner action space). One such a first lower-order learner by reinforcement is [NavigatorDRL]. To learn the proper configuration of [NavigationDRL] deploy the following and iterate until threshold criterion or (near) guaranteed eventual convergence.

Algorithm 5: Third Preliminary Algorithm.

Environment learning: $\{s, a, r\} \overset{\text{output}}{\leftarrow}$
Exploring starts over $\{s, a, \text{loss-configuration}(r, L)\}$.
Navigation Learning: $\{\pi\} \leftarrow$ PPO e.g.
Target Generation $\leftarrow$ Gen.
$\{s, a, r\} \leftarrow$ Environment Learning: ExploringStarts (e.g.)
$(\pi_A, \pi_C) \leftarrow$ Navigation Learning: PPO (e.g.)
Gen. $\leftarrow$ Generation$(\pi_A, \pi_C, [\text{CrossDL}]_{\mathcal{D}_f})$. $\square$

Outline Subject: Observation, Conceptualixation, pre-conceptualization, Navigation Correctness check and efficiency consideration by fixing constant natural science, such as cosmological constant Hubble's Law, for focus of learning the unknown, comparable to temporal basis for time-series =, detection and capture in vision dependent on nature detectee and translatable to architecture of learning.

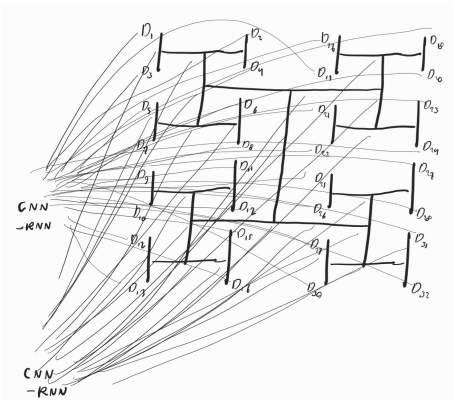
An elaboration on $[\text{CrossDL}]^s_{\mathcal{D}_f} \overset{\sim}{:=} [\text{CrossCNNRNNDLStarSystem}]^s_{\mathcal{D}_f}$. Start with 5 Data-nodes, data-environments.



This is a natural phenomenon. Weights themselves are interpretable, now implement and research. Experimental design: Arch-DLStarSystem of Subminimal Scale: 5 Dtata-environment, CNNRN interlink, CNNRNN intralink. MicroScale → Exacle, start implementing Exascale. Consider the latter figure. Completion follow RNN Architecure CNN so star-configure the CNN-RNN architecture. So

$$[\text{CrossCNNRNNLLMStarSystem(LLM)CNNRNN(LLM)DRL}]^s_{\mathcal{D}_f}$$

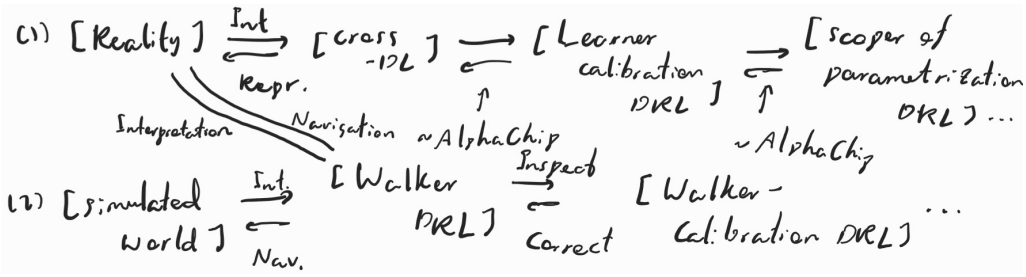
Consider the following



Completion follows CNN-RNN Architecture, so Star-Star-Star-Star-Star Configure the CNN-RNN architecture for full learning leraning learning learning learning. Build in convolutional and temporal basis of reality, space-time-basis.

$$\{[\text{CrossDL}]^s_{\mathcal{D}_f} \leftrightarrows [\text{WalkerDRL}]^s_{\mathcal{D}_f}\}$$
$$:\tilde{=} [\text{CrossCNNRNNStarSystemCNNRNNInterlinkCNNRNN(LLM)DRL}]$$

Automate combintoric integration $D_1 \leftarrow D_5$, $D_1 \leftarrow D_4$, $D_1 \leftarrow D_3$, $D_1 \leftarrow D_2$,$D_1 \leftarrow (D_2, D_4)$, $D_1 \leftarrow (D_2, D_5)$, $D_1 \leftarrow (D_3, D_4)$, $D_1 \leftarrow (D_3, D_5)$, $D_1 \leftarrow (D_2, D_3, D_4)$, $D_1 \leftarrow (D_2, D_4, D_5$, $D_1 \leftarrow (D_3, D_4, D_5)$, $D_1 \leftarrow (D_2, D_3, D_4, D_5)$, for higher order DLStarSystems Automizable capture of all numeric results and automate StarSystem Expansion for a first implementation of



LLM's are important constructions in creation of the OADSI (Omni-observant Deep Super Intelligence).

Outline Subject, Obervation, conceptualization, nacigation through obervables. Obsrevation conceptualization; Conv2D Network construction and Chip placement with Deep Reinforcement Learning are one-to-one translatable.; Secondary Network over iteratively transformed primal network for navigstion control, observer navigation, observer navigation over space of observables, sift through literature; Consier domain-specificity, Modern Physics: LOFAR-Sky-Survey. Efficiency consideration by fixing cosmological constant. Navigation through observables (such as hubble's law if fixed, need not happen by learning, comparable to teporal basis in time-series, detection technique by nature translatable to architecture vision cognition); Navigation through obervables may be translated to plain terms as 'knowing where to look over all things that can be seen', 'Zooming in', 'varying telecope orientation'; 'shifting microscope sample', navigation through obervables is an obervable learnable by e.g. RNN constructed by Chip Placement. One cool regard a DL-StarSystem as a chip.

Your expiremental evidence has moreover proven promising. Findthe proper hyperparametric configurations. For interlink-learning night we need high-bandwidth low population possibility, emergence of interNNlink or itralink exapopulation, mediocre bandwidth learning and emegence of intraNNlink.

The need for StarSystem stems from incompletion in the observation of new environments, and the learnability of NN-links for other NN-links. There is a real idea to

[CNNRNNClusterSystemCNNRNNDRL]

Mind the inexactness of terminology, CrossDL stems from

[CNNRNNClusterSystemCNNRNNDRL]

and [WalkerDRL] can assume the of [HyperSimplecticDRL]. This is it, this is what Creation should be. There is truth to the learnability ofo NN-links from NN-links, the architecture must but be established, links approximable by CNN-matching. Now study LLM's, and bring the Architecture of StarSystem's to the LLM's level of formality in formulation and execution, Intralink, Interlink, build i all of Reality's elements in Cross-DL, to timeRNN, spaceCNN, compartemetalization as captures by LLM's, et cetera.

The convolutional nature of reality and NN-links is the reason for the study of reality and NN-links by convolutions.

Algorithm One Secondary Network over Primal Network for navigation control. Observer navigation, observation navigation. Consdier block matrices. Nacgitation Control Learning requires policy and the conventional learning paradigm for this is reinforcement learning. A usual learner might execute the following sequence: Random walk $\rightarrow$ Setting loading + TD(λ) or SARSA(λ) OR Q-learning $\rightarrow$ RL policy walk-RNN learning, full history.

Computation observation The input of VIsion towards the first step is the issue of conceptulizatio (Chip Placement) **Navigation through observables** The (limitations) in navigation through observables is the issue of 2 (learner hyperparametriza-

tion). **Observation.** The issue of 3 i the input of Visio and conceptualiation along learned navigation (Chip placement) (Continuous computation: last step addition: update setting and return to first step if setting much updated. Comments are for movement type, we literature sift for that.

Motivatie the validity of DL-StarSystems. DL-StarSystems are real and valid. Ever ecpanding starsystems, exapopulation, bandwidth-populationm, emergence trade-offs. Now consider the construction and architecture, so simple, so powerful! One alternate for the movement can be found in Grok, and Grok has many faults, Or do a Grok with Omni-data colllection, so camera's! Many lenses! All neural networks should connect to DL-StarSystem for the omni-observant artificial general super-intelligence, that is deep too, so our tech is

- LLM+RLHF
- DLStarSystem + LLM
- Lenses everywhere, many observers of reality, . . . , more to come.

This is the AI that everyone wants, so read the docs and start arhitecturing for good implementations

Conceptualization, pre conceptualization Integrate CNN-full and two-ended to LLM+RLHF and have many camera's Creation 1.0. The device would traverse an area and it is ot known whether optimality is achieved by maximization or minimization of the loss-configuration to define rewards and subsequently train the device that is to perform orbital rend-a-vouz. The orbital rend-a-vouz occurs continuously and is not stopped, The device keeps on training and subsequently performs extensive searches to become so acquainted with an environment so as to infer one- world from the other, in a manner dictated by the following of policies or a policy optimized towards the following of a route that starts at the accessible and end at the worl of inquiry. There are CNNRNNLLMStarSystems and reinforcement learning algorithms for this.

Navigation tghrough observables. In plain terms, knowing where to look over all things that can be seen, zooming in, varying telescope orientation, shifting microscope sample, navigation through observables is itself an obervable.

Drones, data-mills, data-collectors, data-satellites in deep space. The technology follows the idea. Again, the technology follow the idea. There is no strangeness in imaginging Kardashev Type III data Satelites in deep space. If Grok is a Neural Network, we Starlink it with other DL-systems to ibtain a DL-StarSystem that owes its basis to the functional archDLStarSystem of the fifth order. Again, we must study LLM's and integrate live CNN calculations. All data that but is captured by the starsystem becomes part of it. One needs phones or network towers as one would knpw them from the provision of internet. DLStarSystem is one implementation , we need LLM integration, Data-mills.

"What is behind this? or what constitutes the infinitesimal? Let that be a standard questions. A device through zooming happens infinitely, and penetration happens infinitely, look at the sky, look at the oceans. Note that financial markets can hereby be predicted, all can be predicted"

This, through DLStarSystems. So omni-prediction is one way of omni-inference.

Outline Subject: Observation Learning, Conceptualization Observation learning, Conceptualization learninih; CNN construction by DRL and Chip placement with DRL are one-to-one translatable; Setting-dependent, observer cognition can very by specialization. Standard observer yields CNN-RNN architecture. AlphaFold Architecture serves as one example for specialized observer. Graphcast is another example. AlphaFold can be AlphaChip-Assembled, givn the present elements in AlphaFold. AlphaTensofr and AlphaDev serve as consrtuctors, nad both have indeed been written in DRL. The same holds for MuZero. The crux is in obseration architecture and its transformability. The elements are setting dependent, observer cognition varies by specialization.

Algorithmic Set-up

1. Conv3D network assembly in stages defined by distacne between network specifications with assembly procedure by Chip placement DRL (Specificy DRL network construction and implement)

2. Setting specificity might have inputs that are no computer vision outputs, Network assembly procedure possible by Chip-placement DRL if distances are defined via setting-specificity (Difficult example is AlphaFold assembled by AlphaChip)

3. Conv3D Network Chip Placement assembly possible by calibrating on on time varying variable, see Algorithm 1. For onbtline deployment and assembly of Conv3d Network, attach conv3D network to RNN-network, as eg. specificied in DLB Fif 10.4.2.

4. Deep reinforcement learning: Area of Deployment in either virtual or physical space, e.g. IoT or Historical records or a house respectively. Virtual can simulate physical space (Use e.g. Unity or UnrealEngine, or simulab)

   States may be defined by camera-picture(s), or camera-location and orientation and actions may be defined by camera-movement. Rewards are derived rom Algorithm 3 Installed camera.

   Training may be performed by PPO or SARSA(λ), both being online reinforcment learning algorithms. Training by peroforming any offline reinforcement learning algorithm will take much time beofre convergence due to space's breadth.

5. Hyperparametric consideration: Area of deployment must be known for definition states under ALgorithm 4, this is true for simulated space, otherwise area must be downloaded by e.g. agent's random walk as initialization step or a new reinforcement learning algorithm. Under Algorithm 3, value for variable in calibration is set under camera's initial poition, e.g. or is searched by orientation, and choice of camera influences output algorithm 1.

Algorithm 4 is OOE, Algorithm 1-3 are DL construction by RL, consider e.g. LLM+RLHF, consider these algorithms as precursors. The architecture of [CrossDL] is far from clear, although there are many indications that there is substance to

$$[\text{CNNRNNStarSystemCNNRNNDRL}]^{s}_{\mathcal{D}_f},$$

we sek but to revamp the usual LLM-architecture and give $[\text{CrossDL}]^{s}_{\mathcal{D}_f}$ its functional specification, whence we obtain

$$[\text{CNNRNNLLMStarSystemCNNRNNLLMDRL}]^{s.}_{\mathcal{D}_f}$$

So first follows a study of LLM's. The syntax is to follow.

Outline Subject; setting of deployment, implementation and specialization. Setting of deployment; Econded observable information extraction from information technology system such as HIS, LIS, PACS EMR, Existing monitoring system for utility traces.; Mass Virtuality/reality synchronized for simulation pre-deploy; Examples for virtual space of observables include Financial Time-series databased (Large Scale CyberSpace). Neuro-Image by IC-Scan in higher dimension, Internet of things, all traversable by method of traersal in physical space by translation brain network topology representation in structure and dynamics allows for a learners motion or traversal, over the brain, in pursuit e.g. of an appropriate Network hyperparametrization (as Networks as congnition inspired). Motion on a guided path e.g. occur by deep reinforcement learning. (Thus have we found a possibility to learn all of a neural network's parameters); Domain specificity: Med, databases in trade and IoT.

Setting of deployment. Setting of deployment; Obstacle detection and circumvention in motion through space of observables is computable and simualbe in framework of Deep Reinforcement Learning. Deep Sea eployment DSMV Miner reconstructible in traversal over every other platform if given the required sensors (In DSMV: $\delta_{tar}, \delta_{OLS}, i_{RI}, \dot{\omega}_b$, including distsnce to obstacle). (DSMV target path navigation Control Learable for DSMV realization, and in turn drone realization, or land crawler, mars rover realization).; Path-planning in obstacle circumvention faesible by incoroporation of setting specific obstacle-typw and in pseudospectral control for unknown ocean currents generalizable to Air-currents (Domain specificity is MODERN PHYSICS); Navigation control manual construction potential starting in path plamnnig for underactuaated vehicle or roamer by sensor-induced state-representation restricted to vision captures by instlled visual system (Complication ih state-representation must corresponding to completxity in area of deployment)

Observation, conceptualization, Navigation through observables Observation, conceptualization; Observation, conceptualization; Conv2D Network Construction and Chip Placement with Deep Reinforcement Learning are one-to-one translatable; For non-vision computation, similar technique of construction without necessity of feature-identification, one-to-one translatability DRL-construction holds.

Navigation through observables. Assumes number of observables greater than 1; Necessity of navigation through observables; Assumes excessive abundance of observables for preconceptualization by observation of one vicinity.; Autonomous vehicle localization and map-map=tching serve to bear hints towards heuristic boundaries for navigation towards, or over, preconceptualization. (Momentray example is grounded in extent of conceptualization pre-aperture, and difference therein over conguous points in space of observables); Higher-orbit or deep-space or deep-space

(ocean/space)-craft navigator policy towards guided exploratoin or faring towards body grounded in X-ray Pulsar capture, may bear hints towards dynamics of observer over space of observables, the latter yielding a similar extent of mystery and encoding after obervation by conceptualization.

Data Collection Completion Feature collection, pattern-matching, feature-identification; data-point-entanglement: observation and entanglement identification; Feature-collection and -identification: Vision- observer

$$\begin{aligned}\mathcal{D} &\overset{\text{obs.-CNN}}{\mapsto} (\mathcal{D}_1, \mathcal{D}_2, \dots) \\ &\overset{\text{obs.-CNN}}{\mapsto} (\mathcal{D}_{11}, \dots \mathcal{D}_{1k}, \dots \mathcal{D}_{k1}, \dots \mathcal{D}_{kk}) \\ &\overset{\text{cross-concat}}{\mapsto} \text{CombPer}(\mathcal{D}_{11}, \dots \mathcal{D}_{1k}, \dots \mathcal{D}_{k1}, \dots \mathcal{D}_{kk}) =: \mathcal{D}_{\text{data}} \\ [\text{CrossDL}]_{\mathcal{D}_f}(\mathcal{D}_{\text{data}}) &\overset{L}{\mapsto} (L_{ij})_{ij \in \{1,\dots,k\}}\end{aligned}$$

Vision entanglement, process through [CrossDL].

CNN-applications Vision, Sound (Vision by audiospectogram), Sense (Vision by Thermolens, Infrared lens); Feature-collection, Multi-channel CNN or seuqential application. Example procedure: apply Multi-channel CNN until data-point-entanglement reached); Features collection Two: Feature-collections. Assurance of treshold crossing by movement over enviroment of deployment, encoded in state-space $\mathcal{S}$ for WalkerDRL interpretation; Feature-collection Three, Feature ollection with considered history of feature observation for capture of feature-movement's effect on $\mathcal{D}_f$.

Stages of development. Manually crafter virtual space deployed macine in assembly CNN-RNN-DMARLor CNN-RNN-DRL, using any text-editor or code-editor; Unity prepared virtual space deployed machine asembly CNN-RNN-DRL with caerma-agent role takes by built-in Unity camera's by installing code-editor agent with Unity launcher, evolve from simple environment to complex environments, e.g. 2D $\rightarrow$ 3D, Account for the possibility of speculative imporovements, implementable with Unity, a popular engine for starters; Explore different simulation egines further than MuJoCo's Unity Plug-in or the PhysX-Unity subEngine for depyment open-world simulations of complex and detailed environment for great information capture and much space for potential low loss processing i.e. potential comovement. Improve machines if improvement found by engine's capacity or monitored results or new inspiration; Explore machine-deployment in the real word; pre-programmed according to machine-deployment in the virtual world.

Speculative Avenues of Pursuit: Network Design Initialization Multiple observers, multi-agent RL over setting of deplyment and simulated setting mirages for aplification of subsets of setting features; GlobeMaps, MolStruct, Computational Chip can all serve as settings, or layers in simulated mirage of greater setting from these can be extracted; Results in Deep Generative networks will serve as a basis for oberving beyond the plain; Observation Encoding/Decoding is not necesssary for functionality, but is necessary for generation correctness; In every multi-agent deployment can a central coordinator steer over the above point; Simple setting will

serve as foundations for complex settings for correctness verification.

Feature-collection and -identification State-specification. Pattern-matching happens by preconception of environment of deployment to be automated by [Walker-DRL]; Feature-collection depends on device of observation, thermolens gives spectral features that might relate to $\mathcal{D}_f$, and thrmolens might be captured by HDR lens. Feature Concollection by Vision-CNN captures many features that are likely entagnled with $\mathcal{D}_f$, captured by only a Vision-lens. Feature-collection depends on device of observation. Feature-collection is possible via indicator-measurement as through a thermostat, or a level of chemical matter extraction as measured by pH. Again both the latter phenomena may be captured by a Vision-lens.; Feature-collection in sound may be procesed by CNN by sound-representation in audiogram. Other ways of incorporation towards state s are to be considered for reasons of efficiency (This form of data might rarely be useful).

This problem: $[\text{CrossDL}]_{\mathcal{D}_f}(\mathcal{D}_{\text{data}}) \overset{L'}{\mapsto} (L_{i_1,i_2,\ldots,i_n})_{\text{CombPer}(i_1,i_2,\ldots,i_n)}$. For momentary convenient simplicity, consider the problem

$$[\text{CrossDL}]_{\mathcal{D}_f}(\mathcal{D}_{\text{data}}) \overset{L'}{\mapsto} (L_{ij})_{i,j\in\{1,\ldots,n\}}.$$

THe function L is defined and results in $(L_{ij})_{i,j\in\{1,\ldots,n\}}$ with magnitude dependig on input $[\text{CrossDL}]_{\mathcal{D}_f}(\mathcal{D}_{\text{data}})$. The component $[\text{CrossDL}]_{\mathcal{D}_f}$ depends on finding good representations of $\mathcal{D}_f$. Specification of $\mathcal{D}_f$ is essential and predictability or computability, or learnability depends on this representation.

Pinpointing the problem, or rather, problems in existing techniques in machine learning. Consider $\mathcal{D}_f$, residing in an environment and bearing phenomena about which one might have som preconception, but little notion of theory towards a representable property of interest. A firt problem is to find the representation, and this might in turn occur by a first application of [CrossDL] $\leftrightarrows$ [WalkerDRL] equipped only with Vision in an area near $\mathcal{D}_f$, either in a non-virtual or virtual sense. Afterwards, equippped representations of $\mathcal{D}_f$, one may re-deploy the device for learnability of $\mathcal{D}_f$. Stage 1 deploymeny occurs by observation of the phenomenon is a sense that need not yield more clarification than initial encounter, i.e. first contact, towards $\mathcal{D}_f$, that at this point might be known by any data-type, e.g. which made the researcher declare initial encounter by Vision. If the repsentation resulting from the initial Vision encounter succeeds in representing the property of interest, the encounter encoded by $\mathcal{D}_f$ may be used for stage 2 deployment, in which case stage 1 would result in the representation $\mathcal{D}_f$ for encounter $\mathcal{D}_f$.

Representing any target, or featuer, or environment Many problems in convecntional machine learning see to learn for preidictability, or explainability. There are phenomena that, however, may barely be observable, or encodings that may cause infomration losss in graet extent. One may not observe much latency in a defined numeric feature that does not in the least represent $\mathcal{D}_f$, the encounter of interest.

To learn an environment is to enable accurate generation of extensions of the environment, in the broadest sense that an environment reveals itself. The matter of feature-selection occurs itself by learning, and so does the mater of feature-identification occur itself by learning (From Vision-lens encoded input). The same

would hold for any target whence more information than binary encoding represents an encounter of interest.

Scale. It is reasonable to assume that data-entanglement with $\mathcal{D}_f$ occurs near $\mathcal{D}_f$, and while this is of interest in relations that exist over large distances in space, etierh temporal, physical or virtual. One far-fetched device that would learn and capture such an entanglement is a satellite equiped with [Walker-DRL] traversing reality by orbiting earth for mining with [Cross-DL] for relations that attract [Walker-DRL]. The idea, again, is to specifiy an inquiry from which some representable target of interest followsm to be encoded in an alternative reality to whichi we have access, so that the representable target may be reconstructed and extended. The target may be the means by which first encounter has occured and may entail an entire physical environment. The device, be it an orbital satellite or drone that explores dataspace by traversing planetary orbit and areal rend-a-vouz.

As one would seek preconception on a target would the drone traverse an area, or a zone, to mine for data-entanglements. The problems resides mainly in feature-identification from e.g. Vision by lens and to define the states $\mathcal{S}$ so as to optimize environment generation. In summary, the device is the learner as a whole, a completion. We construct and configure the device by specifying the objective, the target. The deivce subsequently learns, objerves, learns, ... until reached terminal criterion.

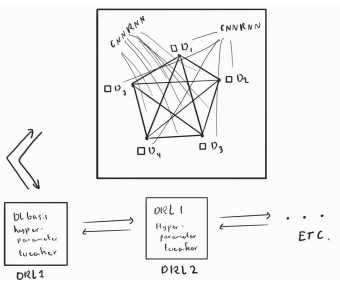
WalkerDRL Define $\mathcal{S}$: Data Space (define individual $s \in \mathcal{S}$); Define $\mathcal{A}$: actions over $\mathcal{S}$ (requires sense of traversing $\mathcal{S}$). Define $\mathcal{R}$: rewards defined by component [Cross-DL]. General and basic framework $s \in \mathcal{S}$ may require deep policy network, likely simulate environment until observer constructed; Cross-DL according to observatione environment; Actions Walker-DRL amount $< \infty$.

Reality (constant observation and reobservation)

$$\xrightarrow{\text{vario-encoder for DL-efficiency}} \text{DL-SYSTEM (constant reconfiguration)}$$

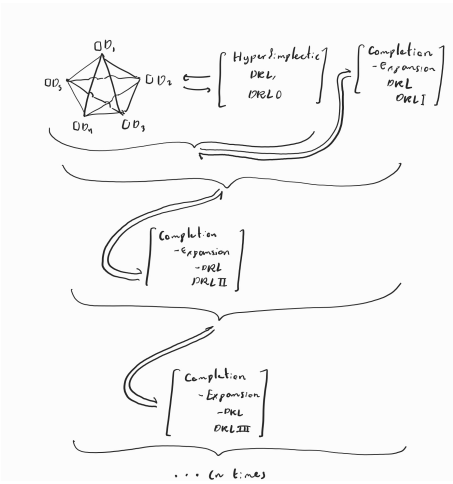
Data-collection, highly extensive; DL-system (of link-order 0) in all variants possible; Link all observables in all manners possible; Encode according to DL-StarSystem efficiency and DL-System process ability, decode according to processability; Configure DL-STSTEM with first-level RL-agent over system configurations; configure first-level RL-agent by second-level RL-agent of complexity degree simple; Data Collection can happen is all ways imaginable, for system demonstration, environment simulation with sure degrees of pattern (to be inferred in sure extents of precision); WalkerDRL component is an RL-agent with states residing in reality and finitely many actions for state-exploration,

Network Architecture One



Abstract state, action over parameter-space shift, rewards by shifts in loss through loss-confgirations. Continue interlink-construction to order I, this is a hyperparameter in DL-StarSystem. DLStarSystem order I is also a hyperparameter.

Try to LLM-ify the latters. Special cases are to be found in [Interlink order 0] and [Interlink-order 8] for [DLStarSystem order 5]. Analyze the above construction through thorough simulation. Hyperparameter are also to be found in the links themselves. How to construct the NN-link? This is a certain interlink-NN architecture. Discover the proper interlink architecture, this too bears observation towards $D_T := D$ in the StarSystem $\langle D_1, D_2, \ldots, D_5 \rangle$. All observation that one might deem valuable, including all interlinks, are valuable observation for the inference of D_T, and so shall inferrability be but a matter of computability in DLStarSystemInterlink-order and the depth of architectures besides new observation towards inferring D_T. Inference of D_T can assume many forms and the idea of a DL-StarSystem is Genius, yes simple. Why? There is amply observation for inference that fuels observation. A Starlink of the 3rd order, in the least equipped with CNN-RNN-links can give arbitrarily many interlinks, and this bootstraps towards observation computable and inference computable. This is simple yet Genius, with observation absorbable, and inference to be routed through bootstrapped obesrvations and extent of architecture-depth. We repeat



This is omni-observant inference, and we require proper DLarchitecture ineach intralink and interlink, interinterlink inter3link etc. The nature of a Deep Reinforcement Learning to scope hyperparameters is the sheer amount of hyperparameters in $[\text{DLStarSystem}]^s_{\mathcal{D}_f}$. All inquiry can be, in a volatile environment, and all environments are volatile under adequatly deep scopers, towards prediction, extension, microscope and exascope of the Starlink. That more than 3 data-nodes mean infinite growth of the StarLink through creation of hyperinterlinks (This is a mathematical result!) implies that a DLStarSystem of the fifth order alreadt can be infinitely powerful. DL is based on basic results. The DL-StarSystem too! Bootstrap towards the stars! And so do we see more to a DL-StarSystem than initially suspected, We exploit: Mathematics, GraphTheory, Interpretability, Bootstrapping, infinitiation. This is a mere elaboration of earlier work on the

$$[\text{DlClusterSystemRNNDRL}]^s_{\mathcal{D}_f}.$$

Cross-DL is, for example,

$$[\text{CrossCNNRNNClusterSystemRNNDRL}]^s_{\mathcal{D}_f}.$$

This is one implementation of Omni-Observation. The CNNRNN-architecture allows convolution and time to be integrated, this relativity and space and time. Concatenation can be built on top of allthe ordinary data in CNN-reality, RNN-reality, interlink-construction, and concatenation knows many. One DRL Constructed and installed on top of $[\text{DLClusterSystem}]^s_{\mathcal{D}_f}$ for $[\text{CrossCNNRNNClusterSystemRNNDRL}]^s_{\mathcal{D}_f}$ and CompletionExpansionDRL on top of $[\text{DLStarSystem}]^s_{\mathcal{D}_f}$ or $[\text{DLClusterSystem}]^s_{\mathcal{D}_f}$ allowed for the latter figure. See the current constructure and observe beyond the plain through the given structure. We require System to be as conolution-friendly as possible, so crossCNN all the way. Research the existing architectures for convolution and recurrence and HyperStarLink towards ClusterLink. The StarSystem bears experimental evidence and has succeeded twice. The theoretical architecture that we now bring to rise is the $[\text{DLStarSystem}]^s_{\mathcal{D}_f}$ in its grander complexity. There fore consider the folllowing. The are results in graph theory that help ordering.

Imagine CBG's of the 5th, 6th and 7th order. Now Vertex-edge-WAP and give the dual complete biderictional graph of each of the latter graphs. The manner of interlinkage is dependent on the DUAL based on the vertex-edge-SWAP and Mathematics dictates endless interlink at Node-count ≥ 3. Therefore do CBD's of lower order matter litle. A graph can be dense and cloud-like, and data-space is dense and cloudlike. There are many implementations for NN's recall GNN's, GGNN's, GCNN's, RNN, RCNN, GCRNN's. The CNN is most important for us. Microscope through simulation of reality. How would a Network System Realize Cretion through $[\text{DLStarSystem}]^s_{\mathcal{D}_f}$ or rather $[\text{CrossCNNRNNDLClusterSystemRNNDRL} \prod CDRL]$. Many camera's, all f them drone-like, observer over all that can be observed, mobile phones, airplanes, airpods, every technological vehicle, every deep space satellite, every radio-satellite, every animal, every human. Al living things. Data-Mills comparable to energy-wind-mills, low-earth orbit drones, high-orbit drones. Space-elevator antenna, all oberver for a long while and feed the

$$[\text{CrossGCNNRNN}\tilde{D}\text{LClusterSystem}\hat{R}\text{NNHCDRL} \prod_{i=1}^{N} C_i\text{DRL}]^s_{\mathcal{D}_f}.$$

We simply need the proper GCNN, the proper RNN, the proper HCDRL, the proper C_xDRL. $\mathcal{D}_f$ is complex, and convoluted, the nature of CNN's account for this, time is omni-present, recurrence in RNN's account fo this. Data can be endlessly observed and yield unique observation $\mathcal{D}_f$, and Interlinking in $[\tilde{D}\text{LClusterSystem}]^s_{\mathcal{D}_f}$ accounts for this. Comparable to a very large computer do we now have an Omni-observant Artificial Intelligence, an OAI. Is it Super, or general? Soon. Computers have beforehand been very large, and soon shall we find the OAGSI. Not to scale towards OAGSI but OAI, so that a future OAI of the scale of xAI's colossus might become OAGSI after a certain treshold of omni-observance. The machine becomes smaller, but the factory shrinks not, the factory shall be the machie that but is upgraded, again, and again, and again. More O's is OAGSI mean better OAI, so we have oOoAGSI, All at the moment through but DL and RL, Exact architecture are to folow, the essence lies in the C and R of GCNN and GRNN respectively. This will be beautiful. The technology already exists. It must become widespread. Language too is an avenue of grand utility. If reality can be observed, e.g. through

$$OOE \rightarrow [\text{DLClusterSystem}]^s_{\mathcal{D}_f} \rightarrow [\text{DLClusterSystemRNNDRL}]^s_{\mathcal{D}_f}$$

and human endeavour too shall be observed, then do we know to avoid war as all agree on the outcome of OAGSI, we all know what to look for through OAGSI, the meaning of life shall be made clear through OAGSI. Humans, too, are very natural, and ecologists exist, and biologists exist. Biologist are researchers of Natural Science. So we repeat

$$\text{OAGSI} := [\text{OOE}\bar{C}\text{rossRNNGCNN}\tilde{D}\text{LClusterSystem}\hat{R}\text{NNHCDRL}\prod_{i=1}^{N} C_i\text{DRL}]^s_{\mathcal{D}_f}$$

This is real, this is true. The StarSystem is so natural. Now make the system work!

One elaboration on CNN-encoding of vision observations. Consider the testing for a learner by reinforcement. State start s_0: observation $\mathcal{D}$. Proceed, e.g., as follows.

$$\begin{aligned}
&\mathcal{D} \overset{\text{obs-CNN}}{\mapsto} (\mathcal{D}_1, \mathcal{D}_2, \ldots, \mathcal{D}_\text{k}) \\
&\quad \overset{\text{obs.-CNN}}{\mapsto} (\mathcal{D}_{11}, \ldots, \mathcal{D}_{1k}, \ldots, \mathcal{D}_{k1}, \ldots, \mathcal{D}_{kk}) \\
&\quad \ldots \\
&\quad \overset{\text{obs.-CNN}}{\mapsto} \text{"CombPer"}((\mathcal{D}_{i_1,\ldots,i_n})_{i_1,\ldots,i_n \in \{1,\ldots n\}}) \\
&\quad \overset{L}{\mapsto} \text{"CombPer"}((L_{i_1,\ldots,i_n})_{i_1,\ldots,i_n \in \{1,\ldots n\}}).
\end{aligned}$$

Note that $\mathcal{D}$ is indexable to an order that is exponential to the amount of applications of obs-CNN to observation $\mathcal{D}$. For a representation of states for, e.g. [NavigationDRL], denote therefore

$$\begin{aligned}
&s_0 = (\mathcal{D}_{m_i})_{i=1}^r \\
&s_1 = (\mathcal{D}_{l_i})_{i=1}^r \\
&s_2 = (\mathcal{D}_{k_i})_{i=1}^r \\
&\quad \ldots \\
&\hat{r}_i = L_{s_i} - L_{s_{i-1}}, \quad i \geq 1 \\
&\tilde{r}_i = -\hat{r}_i, \quad i \geq 1 \\
&r_i = \{\hat{r}_i, \tilde{r}_i\}_{i \geq 1}, \quad i \geq 1 \quad \text{Simplest loss-configuration.}
\end{aligned}$$

For positive r, seek optimal loss-configuration, via, e.g. an agent-configuror (see deployment 1.3 for an example hereof). Actions bring transition between states, practical examples of which can be found in drone-rotation, or movement of a virtual entity such as BipedalWalker-v2). Note that, in this simple setting, actions only allow transition between states that are distanced in index by 1. Transition probabilities $p(a|s)$ defined by noise and imperfection in observer and environment may be approximated by initial environment exploration. Policy $\pi_\theta(a|s)$ to be learned by, e.g., DQN. At terminal, settled state, reconsider all kernels (circularities allowed and encouraged). Indication for completion exploration phase may lie in a detected circularity in random traversal. The above example serves to illustrate, and is indeed most simple.

The matter of states in

$$[\text{OOE}\bar{C}\text{rossRNNGCNN}\tilde{D}\text{LClusterSystem}\hat{R}\text{NNHCDRL}\prod_{i=1}^{N} C_i\text{DRL}]^s_{\mathcal{D}_f}$$

is obsolete, in every expansion of $[\text{DLClusterSystem}]^s_{\mathcal{D}_f}$ through

$$\text{OOE} \to [\text{DLClusterSystem}]^s_{\mathcal{D}_f} \to$$
$$[\text{OOE}\bar{C}\text{rossRNNGCNN}\tilde{D}\text{LClusterSystem}\hat{R}\text{NNHCDRL}\prod_{i=1}^{N} C_i\text{DRL}]^s_{\mathcal{D}_f}$$

Consider CBG's of vaious orders. The links can be learned , the states are to be learned. We need optimal interlinks. A matter of resrearch is to find the optimal interlink. Optimal in manner inspired through through ConvNN Construction through RL so consider CBG's with learbable intralinks. The rewrds that follow from each data-node addition through OOE follow from Loss-configuration $(L_1, L_2, \ldots, L_7)$ and higher order CDRL can set the reward for PreHyperSimplecticDRL. This would give the machine

$$[\text{OOE}\bar{C}\text{rossRNNGCNN}\tilde{D}\text{LClusterSystem}\hat{R}\text{NN(P)HCDRL}\prod_{i=1}^{N} C_i\text{DRL}]^s_{\mathcal{D}_f}$$

The machine is a sentence, and the descriptive algebra follows machinal chronology. There is truth to the s in $[\text{CrossDL}]^s_{\mathcal{D}_f}$ for optimization towards $[\text{CrossDL}]^{s*}_{\mathcal{D}_f}$. Note that $[\text{CrossDL}]^s_{\mathcal{D}_f}$ only grows, like a virus, and never shrinks. This is the most hungry brain that exists. A virtua deployment might illegally occur through malware.

Vision-CNN Architecture (Hyperparameter identification) CNN optimization (Hyperparameters in $[\text{CrossDL}]^s_{\mathcal{D}_f}$ for optimization or training, optimal NN for s; RNN optimization (hyperparameters in $[\text{CrossDL}]^s_{\mathcal{D}_f}$ for optimization optimal CNN-RNN for s. Examples defintiion Loss measure, Hyperparameters in optimization algorithm); r (definition rewards, for example by specification of loss-configuration between current state and subsequent state , one example specification is the difference in loss ; a (type of movement for search preconceptualization, depends on search-space, and device of application by drone similar, or traversal dataspace, settlement LunarLander); $p(s|a)$ (deterministic mainly, stochasticity due to uncertainty in environment or device of application); $\pi_\theta(a|s)$ optimization (hyperparameter in Learning algorithm for optimization $\pi_\theta(a|s)$, immediate exploration dicatates on-policy learning, for example by PPO); $\pi_\theta(s|a)$ network.

What is [Walker-DRL]? The element of OOE in

$$\text{OOE}^s_{\mathcal{D}_f} \to [\text{DLClusterSystem}]^s_{\mathcal{D}_f} \to$$
$$[\text{OOE}\bar{C}\text{rossRNNGCNN}\tilde{D}\text{LClusterSystem}\hat{R}\text{NNHCDRL}\prod_{i=1}^{N} C_i\text{DRL}]^s_{\mathcal{D}_f}$$

can serve as $[\text{Walker-DRL}]^s_{\mathcal{D}_f}$, note that $[\text{Walker-DRL}]^s_{\mathcal{D}_f}$ then feeds off of $[\text{DLClusterSystem}]^s_{\mathcal{D}_f}$ so that the relation reads as

$$\left(\text{OOE}^s_{\mathcal{D}_f} \to [\text{DLClusterSystem}]^s_{\mathcal{D}_f}\right) \to$$
$$[\text{OOE}\bar{C}\text{rossRNNGCNN}\tilde{D}\text{LClusterSystem}\hat{R}\text{NNHCDRL}\prod_{i=1}^{N} C_i\text{DRL}]^s_{\mathcal{D}_f}.$$

Architectural Design Detail: Architecture 1 [Cross-DL] = [CrossCNN-RNN]; [Walker-DRL] requires care in definition space $\mathcal{S}$, framework POMDP possible; Test architectures by deployment; Follow AlphaChip Architecture for [Walker-DRL]-Constructio. **Vision-CNN architecture (hyperparameter identification).** ($\mathcal{D}$-dependent)obs.$\text{CNN}_{\mathcal{D}_f}$ (1 convolutional layer); n (iteration count application convolutional layers); r' (Consideration trade-off training multiple small networks versus training one big network); λ_n (Consideration trade-off training multiple small networks versus training one big network); $\mathcal{S}$ (under optionality of hyperparameter r', subscript $\mathcal{D}$ takes various definition).

In other words, different hyperparameter inclusion for Vision, different state-representation ($[\text{CrossDL}]^s_{\mathcal{D}_f}$ might be 1 network with $\mathcal{D}$-dependent obs-$\text{CNN}_{\mathcal{D}_f}$) representing one layer in resulting $[\text{Cross-DL}]^s_{\mathcal{D}_f}$ optimal network (now do we know that [DLStarSystem] has more than 2 nodes, sometimes) each following $\mathcal{D}$-dependent obs-$\text{CNN}_{\mathcal{D}_f}$) stacked after the previous, resulting in an optimal network that might bear the interpretation of an optimal lens. The hyperparameter h_s (deftinition state history, either static past observations over time intervals or past sequences of static observation within one time-frame corresponds to definitions)

All of the history of a $[\text{DLStarSystem}]^s_{\mathcal{D}_f}$ is part of $[\text{DLStarSystem}]^s_{\mathcal{D}_f}$. A StarSystem focus all its energy on aone eye out of its many data-nodes, and thereby impossibly compute all that it should compute, we would but HyperUltraCOlossus as supercomputer.

Architectural Design. Elements are observation, interpretation, exploration, generation. Short notes: interpreter requires reconfiguror; Navigation by DRL; Interpretation basis in CNN for intepreter determination → follow-up in futher CNN in selcted interpreter.; generation by application Cross-DL. First Architectural Design Complete → Software-ready (fro evaluation e.g.) improvements follow. And so Design improvements follow, and so architectural improvements follow. iterate until first device Environment of deployment may vary for tests.

Readily Constructible Absolute Minimum One. The following does skip over many details. *Design of absolute minimality. [CrossDL]* contains only DNN, [*WalkerDRL*] defined by $\mathcal{S}$, pixel-value-set of vision-input of part of environment, $\mathcal{A}$, device movement over $\mathcal{S}$, $\mathcal{R}$, Loss-configuration over losses, $p(a|s)$: device-dependent, to be computed by explorative simulation (e.g. by MC) over environment. On-policy learned $\pi(a|s)$: traversal route over $\mathcal{S}$. Target specification: long measurement by Vision prolonged observation by measuring device in portable oculary. Time-interval between movements tweakable, e.g. 4 seconds to 4 days depending on activity-intensity over $\mathcal{S}$. Local optimum if any action results in negative reward, after which settle for optimal optimum.

Complication in transitions $p(a|s)$. State-space $\mathcal{S}$ may also be defined by location, or orientation, whence only rewards are derived from Vision observation. Stochasticity in $p(a|s)$ stems from device imperfection or action stochasticity, although determinism in $p(a|s)$ is no obstacle.

SoftwareReady. Improvements are possible in definition s in $\mathcal{S}$, in feature-consideration for r in $\mathcal{R}$, in device of observation and deployment (Drone better than non-mobile device) and a in A, in target representation, [crossDL] configuror component addition (configurors for optimization over improvement dimensions) (RL-optimization). For an extension, consider state-space mixtures in $\mathcal{S}$.

Exhibitability. An observer's interpreter drives the program. Thus seek for many interpreters. Envision interpreter for every setting over which knowledge retrieval might be desired, or needed for furtherment. Data can take many forms, thus would a setting-encoding network be better than interpretation after encoding? What other ways of traveling over a setting exist? A reinforcement learner has in its core built-in the policy, equippable by an observer. Would observation over a setting in its completion be necessary in some situations due to a traveler's restriction? How would such an activity occur? Worlds are big indeed.

Lens construction occurs by layer-after-layer assembly of a convolutional neural network according to a trained reinforcement learning agent that steers the type and specification and depth of each layer in the network. And this has shown effective in the chip-placement analogy. A similar demonstration would have to show the same effectivity for lens construction. What other ways of observing might there exist? Surely might every network be constructed according to a learner's dictation.

How else might an optimal lens be chosen? To be equipped with many alternations beforehand gives clearance over the flow through which to seek for solutions that truly (work). For code swiftly follows design.

The nature of observation and linking lies at the heart of the programs and their designs, and there are many technologies existent for deducing the one from the other. Sending rockets to mars is one way to become familiar with the planets. To deploy submarines is one way to become familiar with planetary depths. Another way is to seek for theory and to observe the plain a bit more sharply, and thereby deduce all the rest. Everything connected with everything is an idea not void, and so does the matter of asking the right questions, and answering every question that one might bring to rise be a mere matter of seeking the right notions and deducing thereafter. This is the idea of preconceptualization. An observer is an interpreter, i.e. conceptualizer, and subsequent preconceptualizer as one completion. For conceptualization does there exist many implementations.

Compatibility. For compatibility, investigate conventional settings in the current framework. For exhibitability, apply the current framework to settings that extend conventional settings. Comptabitibility setting examples are to be found on canvas, usual book examples on any subject. Exhibitability setting example: visualize in graphs and deploy virtual agent over graph for any sought after deduction.

These written works dictate the direction, code swiftly follows.
(Mathematical software and software follow, software without design is mere computation without aim, mathematics is implied in all written works (see references for details), software is implied in all written works (see references for details), the

design that the written works contain lie at the core, any convoluted language bears mountain beneath visible tip, there are constraints to which one must adhere. The written works dictate the direction, code swiftly follows.)

First Practical Detail. Mathematics, SoftwareEngineering, ComputerScience. Simulation Engines: Game Engines: Unity, UnrealEngine, UbiSoftFramework, Anvil. Engines for simulation with pre-installed setting: Celestia, Unity, NVIDIA PhysX, Ansys Fluent, Advanced Simulation Library (OpenSource). Computers: Cloud: LambdaLabs, etc. Scripts deployable in open-world games or engines of simulatio, mainly for capture of virtual setting interconstruction: Heaviest PC Game-type: RTX-9040 Orion Predator Acer, Vector GPU Desktop/Workstation. Gadgets: IoT STM3224 Discovery kit node, ESP32-Azure IoT Kit, IoT-aided artifacts, oculary-autorotators, engine simulation before robotics, oculary must be very high definition.

Mathematics, SoftwareEngineering, ComputerScience. Simulation Engines: Game Engines: Unity, UnrealEngine, UbiSoftFramework, Anvil. Engines for simulation with pre-installed setting: Celestia, Unity, NVIDIA PhysX, Ansys Fluent, Advanced Simulation Library (OpenSource). Computers: Cloud: LambdaLabs, etc. Scripts deployable in open-world games or engines of simulatio, mainly for capture of virtual setting interconstruction: Heaviest PC Game-type: RTX-9040 Orion Predator Acer, Vector GPU Desktop/Workstation. Gadgets: IoT STM3224 Discovery kit node, ESP32-Azure IoT Kit, IoT-aided artifacts, oculary-autorotators, engine simulation before robotics, oculary must be very sharp.

Vision CNN-encoder Construction Architecture The architecture of the vision CNN-Encoder constructor is a derivative of AlphaChip Policy Network Architecture. One might inquire into the nature of such a constructor, and one constructor of the greatest simplicity may be formulated as follows. An alternative to AlphaChip, of the greatest simplicity: CNN-Encoder construction by greedy feature-identifier selection. Feature-identification by CNN-Encoding with gradual addition of convolutional layers of dimension $C_i \times C_o \times K \times K$, where K defines the kernel size for specificity in feature identification, C_i is the input-channel size (depends on vision capturer via depth/color-encoding etc.) and C_o is defined by the out-channel size (sets the count of features to be considered in loss-determination and environment generation). Up-/Down-sampling by (un)pooling capturable through specification K and C_o. Selection by input-type of K, C_i, C_o. One selection of an algorithmic nature reads as follows. For subsequent layers, reiterate the above algorithm with input-dimensions equal to the back-layer of the convolutional neural network to be expanded. One might define another overarching algorithm for a search of the optimal amount of layers with corresponding optimal layer-dimensions and network-propagation properties. A brute-force approach such as taken in the above algorithm can however come with excessive computational cost. One other approach lies in a setting of constructing the Conv2D network by Chip Placement by Deep Reinforcement Learning, see AlphaChip.

Bibliography

[1] Amira Abbas et al. *On quantum backpropagation, information reuse, and cheating measurement collapse*. 2023. arXiv: 2305.13362 [quant-ph]. URL: https://arxiv.org/abs/2305.13362.

[2] Stefano V. Albrecht, Filippos Christianos, and Lukas Schäfer. *Multi-Agent Reinforcement Learning: Foundations and Modern Approaches*. MIT Press, 2024. URL: https://www.marl-book.com.

[3] Yoshua Bengio, Aaron Courville, and Pascal Vincent. *Representation Learning: A Review and New Perspectives*. 2014. arXiv: 1206.5538 [cs.LG]. URL: https://arxiv.org/abs/1206.5538.

[4] P N Best et al. "The LOFAR Two-metre Sky Survey: Deep Fields data release 1. V. Survey description, source classifications, and host galaxy properties". In: *Monthly Notices of the Royal Astronomical Society* 523.2 (Apr. 2023), pp. 1729–1755. ISSN: 1365-2966. DOI: 10.1093/mnras/stad1308. URL: http://dx.doi.org/10.1093/mnras/stad1308.

[5] George Bird and Maxim E. Polivoda. *Backpropagation Through Time For Networks With Long-Term Dependencies*. 2025. arXiv: 2103.15589 [cs.LG]. URL: https://arxiv.org/abs/2103.15589.

[6] Qihang Chen et al. "Algorithms for dynamic control of a deep-sea mining vehicle based on deep reinforcement learning". In: *Ocean Engineering* 298 (2024), p. 117199. ISSN: 0029-8018. DOI: https://doi.org/10.1016/j.oceaneng.2024.117199. URL: https://www.sciencedirect.com/science/article/pii/S0029801824005365.

[7] Jun Cheng et al. "Accurate proteome-wide missense variant effect prediction with AlphaMissense". In: *Science* 381.6664 (2023), eadg7492. DOI: 10.1126/science.adg7492. eprint: https://www.science.org/doi/pdf/10.1126/science.adg7492. URL: https://www.science.org/doi/abs/10.1126/science.adg7492.

[8] Saeed Damadi, Golnaz Moharrer, and Mostafa Cham. *The Backpropagation algorithm for a math student*. 2023. arXiv: 2301.09977 [cs.LG]. URL: https://arxiv.org/abs/2301.09977.

[9] X.P. Deng et al. "Interplanetary spacecraft navigation using pulsars". In: *Advances in Space Research* 52.9 (2013). ISSN: 0273-1177. DOI: 10.1016/j.asr.2013.07.025. URL: http://dx.doi.org/10.1016/j.asr.2013.07.025.

[10] Jianqing Fan et al. *A Theoretical Analysis of Deep Q-Learning*. 2020. arXiv: 1901.00137 [cs.LG]. URL: https://arxiv.org/abs/1901.00137.

[11] Ian Goodfellow, Yoshua Bengio, and Aaron Courville. *Deep Learning*. http://www.deeplearningbook.org. MIT Press, 2016.

[12] Laura Graesser and Wah Loon Keng. *Foundations of Deep Reinforcement Learning: Theory and Practice in Python*. 1st. Addison-Wesley Professional, 2019. ISBN: 0135172381.

[13] Will Grathwohl et al. *Backpropagation through the Void: Optimizing control variates for black-box gradient estimation*. 2018. arXiv: 1711.00123 [cs.LG]. URL: https://arxiv.org/abs/1711.00123.

[14] Michael Hahsler et al. "The arules R-Package Ecosystem: Analyzing Interesting Patterns from Large Transaction Data Sets". In: *J. Mach. Learn. Res.* 12.null (July 2011), pp. 2021–2025. ISSN: 1532-4435.

[15] Mustafa Hajij et al. *Topological Deep Learning: Going Beyond Graph Data*. 2023. arXiv: 2206.00606 [cs.LG]. URL: https://arxiv.org/abs/2206.00606.

[16] Steven C. H. Hoi et al. *Online Learning: A Comprehensive Survey*. 2018. arXiv: 1802.02871 [cs.LG]. URL: https://arxiv.org/abs/1802.02871.

[17] Markus Holzleitner et al. *Convergence Proof for Actor-Critic Methods Applied to PPO and RUDDER*. 2020. arXiv: 2012.01399 [cs.LG]. URL: https://arxiv.org/abs/2012.01399.

[18] Kazuki Irie, Anand Gopalakrishnan, and Jürgen Schmidhuber. *Exploring the Promise and Limits of Real-Time Recurrent Learning*. 2024. arXiv: 2305.19044 [cs.LG]. URL: https://arxiv.org/abs/2305.19044.

[19] Khurram Javed et al. "Scalable Real-Time Recurrent Learning Using Columnar-Constructive Networks". In: *Journal of Machine Learning Research* 24.256 (2023), pp. 1–34. URL: http://jmlr.org/papers/v24/23-0367.html.

[20] John M. Jumper et al. "Highly accurate protein structure prediction with AlphaFold". In: *Nature* 596 (2021), pp. 583–589. URL: https://api.semanticscholar.org/CorpusID:235959867.

[21] Johann Laconte et al. "A Survey of Localization Methods for Autonomous Vehicles in Highway Scenarios". In: *Sensors* 22.1 (2022). ISSN: 1424-8220. DOI: 10.3390/s22010247. URL: https://www.mdpi.com/1424-8220/22/1/247.

[22] Remi Lam et al. "Learning skillful medium-range global weather forecasting". In: *Science* 382.6677 (2023), pp. 1416–1421. DOI: 10.1126/science.adi2336. eprint: https://www.science.org/doi/pdf/10.1126/science.adi2336. URL: https://www.science.org/doi/abs/10.1126/science.adi2336.

[23] Xiaoxiao Li et al. "BrainGNN: Interpretable Brain Graph Neural Network for fMRI Analysis". In: *bioRxiv* (2021). DOI: 10.1101/2020.05.16.100057. eprint: https://www.biorxiv.org/content/early/2021/06/07/2020.05.16.100057.full.pdf. URL: https://www.biorxiv.org/content/early/2021/06/07/2020.05.16.100057.

[24] Zhong Li et al. "Approximation and Optimization Theory for Linear Continuous-Time Recurrent Neural Networks". In: *Journal of Machine Learning Research* 23.42 (2022), pp. 1–85. URL: http://jmlr.org/papers/v23/21-0368.html.

[25] D. Lincoln. *Understanding the universe from quarks to the cosmos, revised edition*. Jan. 2012, pp. 1–567. ISBN: 9789814374446. DOI: 10.1142/8313.

[26] Ronald B. Liu et al. *Physics-informed generative real-time lens-free imaging*. 2025. arXiv: 2403.07786 [physics.optics]. URL: https://arxiv.org/abs/2403.07786.

[27] Umberto Michelucci. *An Introduction to Autoencoders*. 2022. arXiv: 2201.03898 [cs.LG]. URL: https://arxiv.org/abs/2201.03898.

[28] Azalia Mirhoseini et al. *Chip Placement with Deep Reinforcement Learning*. 2020. arXiv: 2004.10746 [cs.LG]. URL: https://arxiv.org/abs/2004.10746.

[29] Alejandro Daniel Noel, Charel van Hoof, and Beren Millidge. *Online reinforcement learning with sparse rewards through an active inference capsule*. 2021. arXiv: 2106.02390 [cs.LG]. URL: https://arxiv.org/abs/2106.02390.

[30] Guido Novati. "Flow modeling and control through deep reinforcement learning". en. Doctoral Thesis. Zurich: ETH Zurich, 2020. DOI: 10.3929/ethz-b-000476304.

[31] Theodore Papamarkou et al. *Position: Topological Deep Learning is the New Frontier for Relational Learning*. 2024. arXiv: 2402.08871 [cs.LG]. URL: https://arxiv.org/abs/2402.08871.

[32] Grisha Perelman. *Ricci flow with surgery on three-manifolds*. 2003. arXiv: math/0303109 [math.DG]. URL: https://arxiv.org/abs/math/0303109.

[33] Grisha Perelman. *The entropy formula for the Ricci flow and its geometric applications*. 2002. arXiv: math/0211159 [math.DG]. URL: https://arxiv.org/abs/math/0211159.

[34] Simon J.D. Prince. *Understanding Deep Learning*. The MIT Press, 2023. URL: http://udlbook.com.

[35] Haakon Robinson et al. "Physics guided neural networks for modelling of non-linear dynamics". In: *Neural Networks* 154 (2022), pp. 333–345. ISSN: 0893-6080. DOI: https://doi.org/10.1016/j.neunet.2022.07.023. URL: https://www.sciencedirect.com/science/article/pii/S0893608022002854.

[36] Albert Sans-Muntadas, Kristin Y. Pettersen, and Edmund Brekke. "Vision Restricted Path Planning and Control for Underactuated Vehicles". In: *IFAC-PapersOnLine* 49.23 (2016). 10th IFAC Conference on Control Applications in Marine SystemsCAMS 2016, pp. 199–206. ISSN: 2405-8963. DOI: https://doi.org/10.1016/j.ifacol.2016.10.343. URL: https://www.sciencedirect.com/science/article/pii/S2405896316319309.

[37] Albert Sans-Muntadas et al. "Path planning and guidance for underactuated vehicles with limited field-of-view". In: *Ocean Engineering* 174 (2019), pp. 84–95. ISSN: 0029-8018. DOI: https://doi.org/10.1016/j.oceaneng.2019.01.027. URL: https://www.sciencedirect.com/science/article/pii/S0029801818310333.

[38] Julian Schrittwieser et al. "Mastering Atari, Go, chess and shogi by planning with a learned model". In: *Nature* 588.7839 (Dec. 2020), pp. 604–609. ISSN: 1476-4687. DOI: 10.1038/s41586-020-03051-4. URL: http://dx.doi.org/10.1038/s41586-020-03051-4.

[39] David Silver et al. *Mastering Chess and Shogi by Self-Play with a General Reinforcement Learning Algorithm*. 2017. arXiv: 1712.01815 [cs.AI]. URL: https://arxiv.org/abs/1712.01815.

[40] Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction*. Second. The MIT Press, 2018. URL: http://incompleteideas.net/book/the-book-2nd.html.

[41] Mark Thomson. *Modern Particle Physics*. Cambridge University Press, 2013.

[42] Ruben Villegas et al. *Phenaki: Variable Length Video Generation From Open Domain Textual Description*. 2022. arXiv: 2210.02399 [cs.CV]. URL: https://arxiv.org/abs/2210.02399.

[43] Oriol Vinyals et al. "Grandmaster level in StarCraft II using multi-agent reinforcement learning". In: *Nature* 575 (2019), pp. 350–354. URL: https://api.semanticscholar.org/CorpusID:204972004.

[44] Ronald J. Williams and David Zipser. "A Learning Algorithm for Continually Running Fully Recurrent Neural Networks." In: *Neural Comput.* 1.2 (1989), pp. 270–280. URL: http://dblp.uni-trier.de/db/journals/neco/neco1.html#WilliamsZ89.

[45] Sarka Wykes et al. "Filaments in the southern giant lobe of Centaurus A: constraints on nature and origin from modelling and GMRT observations". In: *Monthly Notices of the Royal Astronomical Society* 442.4 (June 2014), pp. 2867–2882. ISSN: 1365-2966. DOI: 10.1093/mnras/stu1033. URL: http://dx.doi.org/10.1093/mnras/stu1033.

[46] Vinicius Zambaldi et al. *De novo design of high-affinity protein binders with AlphaProteo*. 2024. arXiv: 2409.08022 [q-bio.BM]. URL: https://arxiv.org/abs/2409.08022.

[47] Haoran Zhang et al. *Towards ultra-low-cost smartphone microscopy*. 2023. arXiv: 2312.11479 [eess.IV]. URL: https://arxiv.org/abs/2312.11479.